



Top 77 CAT Profit, Loss and Interest Questions With Video Solutions

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Questions

Instructions

Answer the following questions based on the following information.

A company purchases components A and B from Germany and USA respectively. A and B form 30% and 50% of the total production cost. Current gain is 20%. Due to change in the international scenario, cost of the German mark increased by 30% and that of USA dollar increased by 22%. Due to market conditions, the selling price cannot be increased by more than 10%.

Question 1

What is the maximum current gain possible?

- A 10%
- B 12.5%
- C 0%
- D 7.5%

 [VIEW SOLUTION](#)

Instructions

DIRECTIONS for questions:

These questions are based on the situation given below:

Recently, Ghosh Babu spent his winter vacation on Kyakya Island. During the vacation, he visited the local casino where he came across a new card game. Two players, using a normal deck of 52 playing cards, play this game. One player is called the Dealer and the other is called the Player. First, the Player picks a card at random from the deck. This is called the base card. The amount in rupees equal to the face value of the base card is called the base amount. The face values of Ace, King, Queen and Jack are ten. For other cards, the face value is the number on the card. Once, the Player picks a card from the deck, the Dealer pays him the base amount. Then the dealer picks a card from the deck and this card is called the top card. If the top card is of the same suit as the base card, the Player pays twice the base amount to the Dealer. If the top card is of the same colour as the base card (but not the same suit) then the Player pays the base amount to the Dealer. If the top card happens to be of a different colour than the base card, the Dealer pays the base amount to the Player. Ghosh Babu played the game 4 times. First time he picked eight of clubs and the Dealer picked queen of clubs. Second time, he picked ten of hearts and the dealer picked two of spades. Next time, Ghosh Babu picked six of diamonds and the dealer picked ace of hearts. Lastly, he picked eight of spades and the dealer picked jack of spades. Answer the following questions based on these four games.

Question 2

The initial money Ghosh Babu had (before the beginning of the game sessions) was Rs. X. At no point did he have to borrow any money. What is the minimum possible value of X?

- A 16
- B 8
- C 100
- D 24

 [VIEW SOLUTION](#)

Instructions

For the following questions answer them individually

Question 3

Instead of a metre scale, a cloth merchant uses a faulty 120 cm scale while buying, but uses a faulty 80 cm scale while selling the same cloth. If he offers a discount of 20%, what is his overall profit percentage?

- A 20%
- B 25%
- C 40%
- D 15%

The video solution shows the following calculations on a whiteboard:

cost price = $\text{Rs. } 100 / 100 \text{ cm}$
 $\text{Rs. } 100 = 100 \text{ cm} \cdot 120 \text{ cm}$
nom $\rightarrow$

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Instructions

Directions for the following two questions: Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of - 3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of - 2.5%, while a fall will entail a return of + 2%.

Question 4

The maximum guaranteed return to Shabnam is

- A 0.25%
- B 0.10%
- C 0.20%
- D 0.15%

E 0.30%

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Directions for the following two questions: Shalini is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of - 2%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of - 2.5%, while a fall will entail a return of + 2%.

Question (1)

The maximum guaranteed return to Shalini is

A) 0.25% B) 0.50%
C) 0.20% D) 0.15%
E) 0.30%

Option A
Option B
Option C
100-A

VIDEO SOLUTION

Instructions

DIRECTIONS for questions:

These questions are based on the situation given below:

Recently, Ghosh Babu spent his winter vacation on Kyakya Island. During the vacation, he visited the local casino where he came across a new card game. Two players, using a normal deck of 52 playing cards, play this game. One player is called the Dealer and the other is called the Player. First, the Player picks a card at random from the deck. This is called the base card. The amount in rupees equal to the face value of the base card is called the base amount. The face values of Ace, King, Queen and Jack are ten. For other cards, the face value is the number on the card. Once, the Player picks a card from the deck, the Dealer pays him the base amount. Then the dealer picks a card from the deck and this card is called the top card. If the top card is of the same suit as the base card, the Player pays twice the base amount to the Dealer. If the top card is of the same colour as the base card (but not the same suit) then the Player pays the base amount to the Dealer. If the top card happens to be of a different colour than the base card, the Dealer pays the base amount to the Player. Ghosh Babu played the game 4 times. First time he picked eight of clubs and the Dealer picked queen of clubs. Second time, he picked ten of hearts and the dealer picked two of spades. Next time, Ghosh Babu picked six of diamonds and the dealer picked ace of hearts. Lastly, he picked eight of spades and the dealer picked jack of spades. Answer the following questions based on these four games.

Question 5

If the final amount of money that Ghosh Babu had with him was Rs. 100, what was the initial amount he had with him?

- A 120
- B 8
- C 4
- D 96

VIEW SOLUTION

Question 6

If Ghosh Babu stopped playing the game when his gain would be maximized, the gain in Rs. would have been

- A 12

- B 20
- C 16
- D 4

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CAT Previous Papers PDF

Instructions

For the following questions answer them individually

Question 7

If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

- A 30%
- B 25%
- C 24%
- D 28%

Handwritten solution for Question 7:

Let CP = 100, SP = 120 (20% profit).
 If 60 toys are sold at 40% discount, then SP = 0.6 × 120 = 72.
 Total SP = 60 × 72 = 4320.
 Total CP = 60 × 100 = 6000.
 Profit = 4320 - 6000 = -1680 (Loss).
 Now, 10 toys are destroyed, so 50 toys are left.
 Let the new discount be x%.
 New SP = 100 × (1 - x/100) = 100 - x.
 Total SP for 50 toys = 50 × (100 - x) = 5000 - 50x.
 Total CP for 50 toys = 50 × 100 = 5000.
 For the same profit (loss), we need:
 5000 - 50x = 4320.
 50x = 5000 - 4320 = 680.
 x = 680 / 50 = 13.6%.

[VIDEO SOLUTION](#)

Question 8

Amal invests Rs 12000 at 8% interest, compounded annually, and Rs 10000 at 6% interest, compounded semi-annually, both investments being for one year. Bimal invests his money at 7.5% simple interest for one year. If Amal and Bimal get the same amount of interest, then the amount, in Rupees, invested by Bimal is

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Amal invests Rs 12000 at 8% interest, compounded annually, and Rs 10000 at 6% interest, compounded semi-annually, both investments being for one year. Bimal invests his money at 7.5% simple interest for one year. If Amal and Bimal get the same amount of interest, then the amount, in Rupees, invested by Bimal is

12000×1.08
 $10000 \times 1.03 \times 1.03$

$P \cdot \frac{R}{100} \cdot T$
 $P \cdot \left(1 + \frac{R}{100}\right)^n$
 $P \cdot \left(1 + \frac{R}{100}\right)^{2n}$



VIDEO SOLUTION

Question 9

A shopkeeper sells two tables, each procured at cost price p , to Amal and Asim at a profit of 20% and at a loss of 20%, respectively. Amal sells his table to Bimal at a profit of 30%, while Asim sells his table to Barun at a loss of 30%. If the amounts paid by Bimal and Barun are x and y , respectively, then $(x - y) / p$ equals

- A 1
- B 1.2
- C 0.50
- D 0.7

cracku

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Shopkeeper

P
 $\downarrow$
 Amal
 $1.2P$
 $\downarrow$
 Bimal
 $1.2P \times 1.3 = 1.56P$

P
 $\downarrow$
 Asim
 $0.8P$
 $\downarrow$
 Barun
 $0.8P \times 0.7 = 0.56P$



VIDEO SOLUTION



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Question 10

Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a loss of 25%. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was

- A 6000

- B 8000
- C 4000
- D 2000

cracku

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A) 6000
B) 8000
C) 4000
D) 2000

Handwritten solution: $10 \cdot C$, $6C$, $4C$, 1.25 , $0.75C$, 2000



VIDEO SOLUTION

Question 11

Nitu has an initial capital of ₹20,000. Out of this, she invests ₹8,000 at 5.5% in bank A, ₹5,000 at 5.6% in bank B and the remaining amount at $x\%$ in bank C, each rate being simple interest per annum. Her combined annual interest income from these investments is equal to 5% of the initial capital. If she had invested her entire initial capital in bank C alone, then her annual interest income, in rupees, would have been

- A 700
- B 800
- C 900
- D 1000

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A) 700
B) 800
C) 900

Handwritten solution: $8000 \times 5.5\% + 5000 \times 5.6\% + 7000 \times x\% = 20000 \times 5\%$



VIDEO SOLUTION

Question 12

Anil invests Rs. 22000 for 6 years in a certain scheme with 4% interest per annum, compounded half-yearly. Sunil invests in the same scheme for 5 years, and then reinvests the entire amount received at the end of 5 years for one year at 10% simple interest. If the amounts received by both at the end of 6 years are same, then the initial investment made by Sunil, in rupees, is

Anil invests Rs. 22000 for 6 years in a certain scheme with 4% interest per annum, compounded half-yearly. Sunil invests in the same scheme for 5 years, and then reinvests the entire amount received at the end of 5 years for one year at 10% simple interest. If the amounts received by both at the end of 6 years are same, then the initial investment made by Sunil, in rupees, is

Answer: _____

Handwritten notes on the screen show the calculation for Anil's investment: $22000 \times \left(1 + \frac{4}{100}\right)^{12}$. To the right, the formula for Sunil's investment is shown: $P \left(1 + \frac{R}{100}\right)^N$ and $P \left(1 + \frac{R/2}{100}\right)^{2N}$.

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Question 13

Amal purchases some pens at ₹ 8 each. To sell these, he hires an employee at a fixed wage. He sells 100 of these pens at ₹ 12 each. If the remaining pens are sold at ₹ 11 each, then he makes a net profit of ₹ 300, while he makes a net loss of ₹ 300 if the remaining pens are sold at ₹ 9 each. The wage of the employee, in INR, is

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Handwritten notes on the screen show the calculation: $\text{No. of pens} = N$; $\text{TCP} = 8N + F$. The total selling price (TSP) for the first case is $\text{TSP}_1 = 12 \times 100 + (N - 100) \times 11$. The total selling price (TSP) for the second case is $\text{TSP}_2 = 12 \times 100 + (N - 100) \times 9$.

VIDEO SOLUTION

Question 14

Gopal borrows Rs. X from Ankit at 8% annual interest. He then adds Rs. Y of his own money and lends Rs. X+Y to Ishan at 10% annual interest. At the end of the year, after returning Ankit's dues, the net interest retained by Gopal is the same as that accrued to Ankit. On the other hand, had Gopal lent Rs. X+2Y to Ishan at 10%, then the net interest retained by him would have increased by Rs. 150. If all interests are compounded annually, then find the value of X + Y.

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Handwritten notes on the screen show the following calculations:

$$\text{Ankit: } X \times \frac{8}{100}$$

$$\text{Gopal: } (X+Y) \times \frac{10}{100} - \frac{8X}{100}$$

$$\text{Gopal: } (X+2Y) \times \frac{10}{100} - \frac{8X}{100}$$

A large play button icon is overlaid on the video frame.

VIDEO SOLUTION

Question 15

The income of Amala is 20% more than that of Bimala and 20% less than that of Kamala. If Kamala's income goes down by 4% and Bimala's goes up by 10%, then the percentage by which Kamala's income would exceed Bimala's is nearest to

- A 31
- B 29
- C 28
- D 32

cracku

The income of Amala is 20% more than that of Bimala and 20% less than that of Kamala. If Kamala's income goes down by 4% and Bimala's goes up by 10%, then the percentage by which Kamala's income would exceed Bimala's is nearest to

Handwritten notes on the screen show the following calculations:

$$A = 120$$

$$B = 100$$

$$K = 150$$

$$K \downarrow 4\% = 146$$

$$B \uparrow 10\% = 110$$

A large play button icon is overlaid on the video frame.

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Question 16

I sold two watches for Rs. 300 each, one at the loss of 10% and the other at the profit of 10%. What is the percentage of loss(-) or profit(+) that resulted from the transaction?

- A (+)10
- B (-)1

C (+)1

D (-)10

cracku

I sold two watches for Rs. 300 each, one at the loss of 10% and the other at the profit of 10%. What is the percentage of loss(-) or profit(+) that resulted from the transaction?

A) (+)10
B) (-)1
C) (+)1
D) (-)10

Total SP = $300 + 300 = 600$

Total CP = $\frac{300}{0.9} + \frac{300}{1.1} = 300 + 300 = 600$

10%

VIDEO SOLUTION

Question 17

In a market, the price of medium quality mangoes is half that of good mangoes. A shopkeeper buys 80 kg good mangoes and 40 kg medium quality mangoes from the market and then sells all these at a common price which is 10% less than the price at which he bought the good ones. His overall profit is

A 6%

B 8%

C 10%

D 12%

cracku

In a market, the price of medium quality mangoes is half that of good mangoes. A shopkeeper buys 80 kg good mangoes and 40 kg medium quality mangoes from the market and then sells all these at a common price which is 10% less than the price at which he bought the good ones. His overall profit is

10%

Rs. 1.8/kg

80 kg

40 kg

1.8

VIDEO SOLUTION

Question 18

Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

A 50

B 60

C 54

D 55

cracku

Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

$$0.8M \times 0.8 = 0.64M$$
$$0.4M \times 0.75 = 0.3M$$
$$0.64M + 0.3M = 0.94M$$
$$\frac{0.94M}{0.8M} = 1.175$$
$$17.5\%$$

0.8M x 0.75 = 0.6M

6.4M + 2.4M = 8.8M

M = 2112

8.8

240

88.8



VIDEO SOLUTION

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Instructions

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 66.67%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

Question 19

How many units should Mr. David produce daily?

- A 130
- B 100
- C 70
- D 150
- E Cannot be determined

cracku

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 66.67%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

Num 0 $SP = 30N$

$CP = cN^2 + bN + 240$

$\frac{CP_{40}}{CP_{20}} = \frac{5}{3}$

$\frac{CP_{60}}{CP_{40}} = \frac{3}{2}$

$100c + 20b + 240$



Instructions

For the following questions answer them individually

Question 20

Meena scores 40% in an examination and after review, even though her score is increased by 50%, she fails by 35 marks. If her post-review score is increased by 20%, she will have 7 marks more than the passing score. The percentage score needed for passing the examination is

- A 60
- B 80
- C 70
- D 75

cracku

Meena scores 40% in an examination and after review, even though her score is increased by 50%, she fails by 35 marks. If her post-review score is increased by 20%, she will have 7 marks more than the passing score. The percentage score needed for passing the examination is

A) 60 B) 80
C) 70 D) 75

$$0.40a \times 1.5 = 0.6a$$

$$0.6a \times 1.2 = 0.72a$$

$$0.6a + 35 = \text{passing}$$

$$0.72a = \text{passing} + 7$$



Question 21

A person spent Rs 50000 to purchase a desktop computer and a laptop computer. He sold the desktop at 20% profit and the laptop at 10% loss. If overall he made a 2% profit then the purchase price, in rupees, of the desktop is

A person spent Rs 50000 to purchase a desktop computer and a laptop computer. He sold the desktop at 20% profit and the laptop at 10% loss. If overall he made a 2% profit then the purchase price, in rupees, of the desktop is

$$D + L = 50000$$

$$1.2D + 0.9L = 51000$$

$$50000 \times \frac{102}{100}$$





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Question 22

In a group of people, 28% of the members are young while the rest are old. If 65% of the members are literates, and 25% of the literates are young, then the percentage of old people among the illiterates is nearest to

- A 62
- B 55
- C 59
- D 66

In a group of people, 28% of the members are young while the rest are old. If 65% of the members are literates, and 25% of the literates are young, then the percentage of old people among the illiterates is nearest to

A) 62 B) 55
C) 59 D) 66

Total = 10000
Young = 2800
Old = 7200
Literates = 6500
Illiterates = 3500
Young = 6500 * 0.25 = 1625
Old = 3500 - 1625 = 1875
Percentage of old people among the illiterates = $\frac{1875}{3500} \times 100 = 53.57\%$

VIDEO SOLUTION

Question 23

Veeru invested Rs 10000 at 5% simple annual interest, and exactly after two years, Joy invested Rs 8000 at 10% simple annual interest. How many years after Veeru's investment, will their balances, i.e., principal plus accumulated interest, be equal?

Veeru invested Rs 10000 at 5% simple annual interest, and exactly after two years, Joy invested Rs 8000 at 10% simple annual interest. How many years after Veeru's investment, will their balances, i.e., principal plus accumulated interest, be equal?

$T \rightarrow 10000 \text{ at } 5\%$
 $T-2 \rightarrow 8000 \text{ at } 10\%$
 $\frac{5}{4} \times 10000 \left(1 + \frac{T}{100}\right) = 8000 \left(1 + \frac{10 \times (T-2)}{100}\right)$

VIDEO SOLUTION

Question 24

In the final examination, Bishnu scored 52% and Asha scored 64%. The marks obtained by Bishnu is 23 less, and that by Asha is 34 more than the marks obtained by Ramesh. The marks obtained by Geeta, who scored 84%, is

- A 357
- B 417
- C 439
- D 399

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In the final examination, Bishnu scored 52% and Asha scored 64%. The marks obtained by Bishnu is 23 less, and that by Asha is 34 more than the marks obtained by Ramesh. The marks obtained by Geeta, who scored 84%, is

total marks = 100x
 Bishnu = 52x = R - 23
 Asha = 64x = R + 34

R = 23 + 52x
 R = 34 + 64x

23 + 52x = 34 + 64x
 12x = -11
 x = -11/12

Geeta's marks = 84x = 84 * (-11/12) = -77

77

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Question 25

A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to

- A 22
- B 35
- C 25
- D 31

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A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to

CP of 1 kg of sugar = Rs. 100
 Total CP = 3500
 SP = 120 x 0.9 = 108
 5 x 108 = 540
 15 x 108 = 1620
 3 x 0 = 0
 12x

MP = Rs. 120
 5 x 120 = 600
 15 x 108 = 1620
 3 x 0 = 0

12x

VIDEO SOLUTION

Question 26

Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

Which of the following cannot be true?

- A At least two horses finished before Spotted
- B Red finished last
- C There were three horses between Black and Spotted
- D There were three horses between White and Red
- E Grey came in second

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Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

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 B) Red finished last
 C) There were three horses between Black and Spotted
 D) There were three horses between White and Red
 E) Grey came in second

Handwritten notes on the screen:

1 → 4x
 2 → 3x
 3 → 2x
 4 → 0
 5 → 0

3000 → Red
 2000 → White
 1000 → Black
 6000
 6000

1 →
 2 →
 3 → Red (3000)
 4 →
 5 →



VIDEO SOLUTION

Question 27

Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

Suppose, in addition, it is known that Grey came in fourth. Then which of the following cannot be true?

- A Spotted came in first
- B Red finished last
- C White came in second
- D Black came in second
- E There was one horse between Black and White

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Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

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A) At least two horses finished before Spotted
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 C) There were three horses between Black and Spotted
 D) There were three horses between White and Red
 E) Grey came in second

Handwritten notes on the screen show a list of horses and their positions: 1 → Red, 2 → White, 3 → Grey, 4 → Black, 5 → Spotted. A large play button is overlaid on the video.

VIDEO SOLUTION

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Question 28

A sum of money compounded annually becomes Rs.625 in two years and Rs.675 in three years. The rate of interest per annum is

- A 7%
- B 8%
- C 6%
- D 5%

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A sum of money compounded annually becomes Rs.625 in two years and Rs.675 in three years. The rate of interest per annum is

A) 7%
 B) 8%
 C) 6%
 D) 5%

Handwritten notes on the screen show the formula for compound interest: $A = P(1+r)^n$. A large play button is overlaid on the video.

VIDEO SOLUTION

Question 29

A yearly payment to the servant is Rs. 90 plus one turban. The servant leaves the job after 9 months and receives Rs. 65 and a turban. Then find the price of the turban.

- A Rs. 10
- B Rs. 15
- C Rs. 7.50
- D Cannot be determined

A yearly payment to the servant is Rs. 90 plus one turban. The servant leaves the job after 9 months and receives Rs. 65 and a turban. Then find the price of the turban.

A) Rs. 10
B) Rs. 15
C) Rs. 7.50
D) Cannot be determined

$90 + T \times \frac{9}{12} = 65 + T$

VIDEO SOLUTION

Question 30

Raj invested ₹ 10000 in a fund. At the end of first year, he incurred a loss but his balance was more than ₹ 5000. This balance, when invested for another year, grew and the percentage of growth in the second year was five times the percentage of loss in the first year. If the gain of Raj from the initial investment over the two year period is 35%, then the percentage of loss in the first year is

- A 5
- B 15
- C 17
- D 10

Raj invested ₹ 10000 in a fund. At the end of first year, he incurred a loss but his balance was more than ₹ 5000. This balance, when invested for another year, grew and the percentage of growth in the second year was five times the percentage of loss in the first year. If the gain of Raj from the initial investment over the two year period is 35%, then the percentage of loss in the first year is

$10000 \times (1 - \frac{x}{100}) \times (1 + 5 \times \frac{x}{100}) = 13500$

$7500 < X < 9000$

VIDEO SOLUTION

Free CAT Preparation Video Lectures

Question 31

Anil, Bobby, and Chintu jointly invest in a business and agree to share the overall profit in proportion to their investments. Anil's share of investment is 70%. His share of profit decreases by ₹ 420 if the overall profit goes down from 18% to 15%. Chintu's share of profit increases by ₹ 80 if the overall profit goes up from 15% to 17%. The amount, in INR, invested by Bobby is

- A 2000
- B 2400

C 2200

D 1800

cracku

Anil, Bobby, and Chintu jointly invest in a business and agree to share the overall profit in proportion to their investments. Anil's share of investment is 20%. His share of profit decreases by ₹ 420 if the overall profit goes down from 18% to 15%. Chintu's share of profit increases by ₹ 80 if the overall profit goes up from 15% to 17%. The amount, in INR, invested by Bobby is

Anil = ₹ x
Bobby =
Chintu =

$20\% \times 18\% = 3.6\%$
 $20\% \times 15\% = 3.0\%$
 $3.6\% - 3.0\% = 0.6\%$
 $0.6\% \times x = 420$
 $x = 70000$

Total Investment = $100\% \times x = 100x$
 $x = 20000$

cracku

VIDEO SOLUTION

Question 32

Suppose, C1, C2, C3, C4, and C5 are five companies. The profits made by C1, C2, and C3 are in the ratio 9 : 10 : 8 while the profits made by C2, C4, and C5 are in the ratio 18 : 19 : 20. If C5 has made a profit of Rs 19 crore more than C1, then the total profit (in Rs) made by all five companies is

A 438 crore

B 435 crore

C 348 crore

D 345 crore

cracku

Suppose, C1, C2, C3, C4, and C5 are five companies. The profits made by C1, C2, and C3 are in the ratio 9 : 10 : 8 while the profits made by C2, C4, and C5 are in the ratio 18 : 19 : 20. If C5 has made a profit of Rs 19 crore more than C1, then the total profit (in Rs) made by all five companies is

$9 \times C_1 : C_2 : C_3 = 9 : 10 : 8$
 $5 \times C_2 : C_4 : C_5 = 18 : 19 : 20$
 $C_1 : C_2 : C_3 = 9 : 10 : 8$
 $C_2 : C_4 : C_5 = 18 : 19 : 20$
 $C_1 : C_2 : C_3 : C_4 : C_5 = 9 : 10 : 8 : 19 : 20$

cracku

VIDEO SOLUTION

Question 33

Mayank buys some candies for Rs 15 a dozen and an equal number of different candies for Rs 12 a dozen. He sells all for Rs 16.50 a dozen and makes a profit of Rs 150. How many dozens of candies did he buy altogether?

A 50

B 30

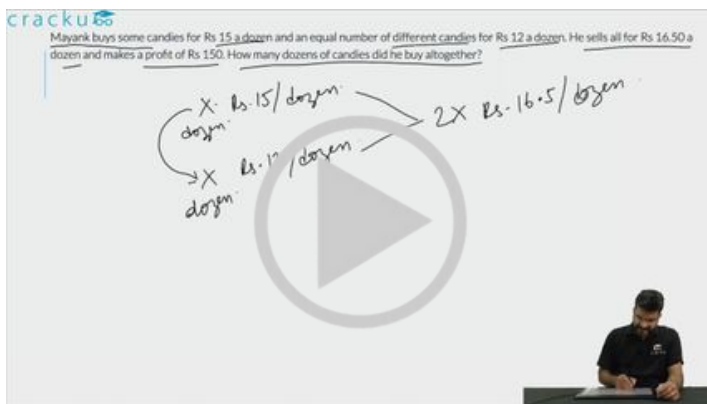
C 25

D 45

cracku

Mayank buys some candies for Rs 15 a dozen and an equal number of different candies for Rs 12 a dozen. He sells all for Rs 16.50 a dozen and makes a profit of Rs 150. How many dozens of candies did he buy altogether?

$X \cdot \text{Rs. } 15/\text{dozen}$
 $X \cdot \text{Rs. } 12/\text{dozen}$
 $2X \cdot \text{Rs. } 16.5/\text{dozen}$



VIDEO SOLUTION

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Question 34

Amal buys 110 kg of syrup and 120 kg of juice, syrup being 20% less costly than juice, per kg. He sells 10 kg of syrup at 10% profit and 20 kg of juice at 20% profit. Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg and makes an overall profit of 64%. Then, Amal's cost price for syrup, in rupees per kg, is

cracku

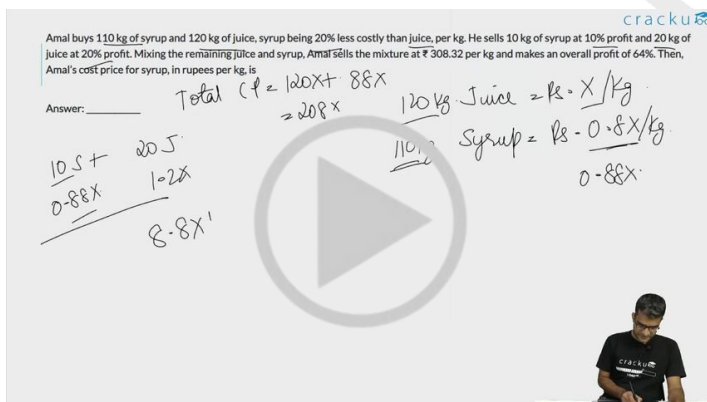
Amal buys 110 kg of syrup and 120 kg of juice, syrup being 20% less costly than juice, per kg. He sells 10 kg of syrup at 10% profit and 20 kg of juice at 20% profit. Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg and makes an overall profit of 64%. Then, Amal's cost price for syrup, in rupees per kg, is

Answer: _____

Total CP = $120X + 88X = 208X$

120 kg Juice = $\text{Rs. } X/\text{kg}$
110 kg Syrup = $\text{Rs. } 0.8X/\text{kg}$

$\frac{10.5 + 20.5}{0.88X} = \frac{10.5}{0.88X} + \frac{20.5}{0.88X}$
 $\frac{31}{0.88X}$



VIDEO SOLUTION

Question 35

Gita sells two objects A and B at the same price such that she makes a profit of 20% on object A and a loss of 10% on object B. If she increases the selling price such that objects A and B are still sold at an equal price and a profit of 10% is made on object B, then the profit made on object A will be nearest to

A 42%

B 45%

C 47%

D 49%

cracku

Gita sells two objects A and B at the same price such that she makes a profit of 20% on object A and a loss of 10% on object B. If she increases the selling price such that objects A and B are still sold at an equal price and a profit of 10% is made on object B, then the profit made on object A will be nearest to

A) 42%
C) 47%

Handwritten notes:

For Object A:
 $SP = X$
 $CP = \frac{X}{1.2}$
 $NSP = \frac{1.1X}{0.9}$

For Object B:
 $SP = X$
 $CP = \frac{X}{0.9}$
 $NSP = \frac{1.1X}{0.9}$

Options: A) 42%, B) 45%, C) 47%

VIDEO SOLUTION

Question 36

The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original selling price, in rupees, of the product is

- A 15
- B 14
- C 10
- D 20

cracku

The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original selling price, in rupees, of the product is

A) 15
C) 10

Handwritten notes:

Selling Price = $1.4X$
 Cost Price = X
 New CP = $0.6X$
 New SP = $1.4X - 5$

Options: A) 15, B) 14, C) 10, D) 20

VIDEO SOLUTION

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Question 37

An amount of Rs 10000 is deposited in bank A for a certain number of years at a simple interest of 5% per annum. On maturity, the total amount received is deposited in bank B for another 5 years at a simple interest of 6% per annum. If the interests received from bank A and bank B are in the ratio 10 : 13, then the investment period, in years, in bank A is

- A 4
- B 5

C 3

D 6

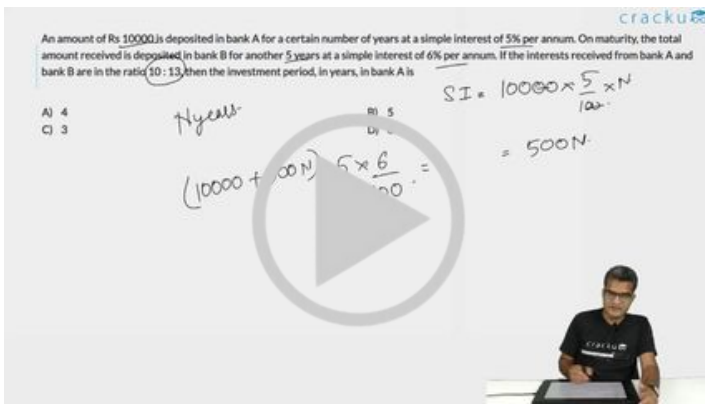
An amount of Rs 10000 is deposited in bank A for a certain number of years at a simple interest of 5% per annum. On maturity, the total amount received is deposited in bank B for another 5 years at a simple interest of 6% per annum. If the interests received from bank A and bank B are in the ratio 10 : 13, then the investment period, in years, in bank A is

A) 4
C) 3

Handwritten solution:

$$SI = 10000 \times \frac{5}{100} \times N = 500N$$
$$(10000 + 500N) \times \frac{5 \times 6}{100} = 500N$$

Years



VIDEO SOLUTION

Question 38

Bina incurs 19% loss when she sells a product at Rs. 4860 to Shyam, who in turn sells this product to Hari. If Bina would have sold this product to Shyam at the purchase price of Hari, she would have obtained 17% profit. Then, the profit, in rupees, made by Shyam is

Bina incurs 19% loss when she sells a product at Rs. 4860 to Shyam, who in turn sells this product to Hari. If Bina would have sold this product to Shyam at the purchase price of Hari, she would have obtained 17% profit. Then, the profit, in rupees, made by Shyam is

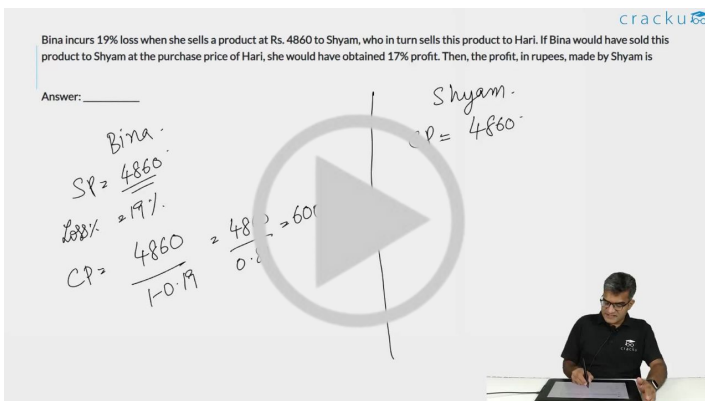
Answer: _____

Handwritten solution:

Bina:

$$SP = 4860$$
$$Loss = 19\%$$
$$CP = \frac{4860}{1 - 0.19} = \frac{4860}{0.81} = 6000$$

Shyam:

$$CP = 4860$$


VIDEO SOLUTION

Question 39

Anil invests Rs 22000 for 6 years in a scheme with 4% interest per annum, compounded half-yearly. Separately, Sunil invests a certain amount in the same scheme for 5 years, and then reinvests the entire amount he receives at the end of 5 years, for one year at 10% simple interest. If the amounts received by both at the end of 6 years are equal, then the initial investment, in rupees, made by Sunil is

A 20860

B 20640

C 20480

D 20808

Anil invests Rs 22000 for 6 years in a scheme with 4% interest per annum, compounded half-yearly. Separately, Sugil invests a certain amount in the same scheme for 5 years, and then reinvests the entire amount he receives at the end of 5 years, for one year at 10% simple interest. If the amounts received by both at the end of 6 years are equal, then the initial investment, in rupees, made by Sunil is

A) 20860
C) 20480
R) 20640
D) 20500

$$22000 \times (1.02)^{12} = P \times (1.02)^{10}$$

$$A = P \left(1 + \frac{R}{2}\right)^{2N}$$

$$A = P \left(1 + \frac{R}{2}\right)^{2N}$$


VIDEO SOLUTION

CAT Percentile Predictor

Question 40

Gopi marks a price on a product in order to make 20% profit. Ravi gets 10% discount on this marked price, and thus saves Rs 15. Then, the profit, in rupees, made by Gopi by selling the product to Ravi, is

- A 10
- B 25
- C 15
- D 20

Gopi marks a price on a product in order to make 20% profit. Ravi gets 10% discount on this marked price, and thus saves Rs 15. Then, the profit, in rupees, made by Gopi by selling the product to Ravi, is

A) 10
C) 15
B) 25
D) 20

Cost Price = C
Marked Price = 1.2C
Selling Price = 0.9C

0.1



VIDEO SOLUTION

Question 41

After two successive increments, Gopal's salary became 187.5% of his initial salary. If the percentage of salary increase in the second increment was twice of that in the first increment, then the percentage of salary increase in the first increment was

- A 30
- B 27.5
- C 25

After two successive increments, Gopal's salary became 187.5% of his initial salary. If the percentage of salary increase in the second increment was twice that in the first increment, then the percentage of salary increase in the first increment was

A) 30
B) 27.5
C) 25
D) 20

X $2X$ $1.875 = \frac{15}{8}$

$S(1+X)(1+2X)$ $S \times 1.875$

VIDEO SOLUTION

Question 42

Jayant bought a certain number of white shirts at the rate of Rs 1000 per piece and a certain number of blue shirts at the rate of Rs 1125 per piece. For each shirt, he then set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10% and made a total profit of Rs.51000. If he bought both colors of shirts, then the maximum possible total number of shirts that he could have bought is

Jayant bought a certain number of white shirts at the rate of Rs 1000 per piece and a certain number of blue shirts at the rate of Rs 1125 per piece. For each shirt, he then set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10% and made a total profit of Rs.51000. If he bought both colors of shirts, then the maximum possible total number of shirts that he could have bought is

Answer: _____

$W \rightarrow 1000$ $B \rightarrow 1125$ $A = \frac{1000W + 1125B}{B+W}$

MP $= 1.25A$
SP $= 1.25A \times 0.9 = 1.125A$

Profit $= 0.125A \times \frac{B+W}{8} = 1000W + 1125B$

VIDEO SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

Instructions

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakhs to buy 100 pieces. He can purchase a colour TV at Rs.10,000 and a VCR at Rs.15,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,500. His objective is to maximize profits. Assume that he can sell all the items that he stocks.

Question 43

The maximum profit, in rupees lakh, the dealer can earn from his original stock if he can sell a colour TV at Rs. 12200 and VCR at Rs.18300 is

- A 2.64
B 2.49
C 2.72

D 2.87

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakhs to buy 100 pieces. He can purchase a colour TV at Rs.15,000 and a VCR at Rs.11,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,300. His objective is to maximize profit. Assume that he can sell all the items that he stocks. The maximum profit, in rupees lakh, the dealer can earn from his original stock if he can sell a colour TV at Rs. 12,000 and VCR at Rs.18,300 is

A) 2.64
B) 2.72
C) 2.49
D) 2.87

Handwritten notes on the screen:

$$a + b = 100$$

$$15a + 11b \leq 1200$$

$$2a + 3b \leq 20$$

Profit calculation:

Item	Cost Price (CP)	Selling Price (SP)	Profit
TV	15k	18.3k	3.3k
VCR	11k	12k	2.2k

VIDEO SOLUTION

Question 44

For the maximum profit, the number of colour TVs and VCRs that he should respectively stock are

- A 80, 20
- B 20, 80
- C 60, 40
- D None of these

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakhs to buy 100 pieces. He can purchase a colour TV at Rs.15,000 and a VCR at Rs.11,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,300. His objective is to maximize profit. Assume that he can sell all the items that he stocks. For the maximum profit, the number of colour TVs and VCRs that he should respectively stock are

A) 80, 20
B) 20, 80
C) None of these

Handwritten notes on the screen:

$$a + b = 100$$

$$15a + 11b \leq 1200$$

$$2a + 3b \leq 20$$

$$2a + 4b + b \leq$$

Profit calculation:

Item	CP	SP	Profit
TV	15k	17.5k	2.5k
VCR	11k	12k	2k

VIDEO SOLUTION

Instructions

For the following questions answer them individually

Question 45

Mr. Pinto invests one-fifth of his capital at 6%, one-third at 10% and the remaining at 1%, each rate being simple interest per annum. Then, the minimum number of years required for the cumulative interest income from these investments to equal or exceed his initial capital is

Mr. Pinto invests one-fifth of his capital at 6%, one-third at 10% and the remaining at 1%, each rate being simple interest per annum. Then, the minimum number of years required for the cumulative interest income from these investments to equal or exceed his initial capital is

Answer: _____

Handwritten notes:

- $\frac{1}{5} \rightarrow 6\%$
- $\frac{1}{3} \rightarrow 10\%$
- 2/5 remain $\rightarrow 1\%$
- Total $= 30X$
- $6X \rightarrow 6\% \rightarrow$
- $10X \rightarrow 10\% \rightarrow$
- $14X \rightarrow 1\% \rightarrow$




VIDEO SOLUTION

Data Interpretation for CAT Questions (download pdf)

Question 46

Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. He repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year. Then, the total interest, in rupees, paid over the three years is nearest to

- A 45311
- B 51311
- C 33130
- D 40991

Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. He repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year. Then, the total interest, in rupees, paid over the three years is nearest to

Options:

- A) 45311
- B) 51311
- C) 33130
- D) 40991

Handwritten notes:

- $200000 \times 1.04 = 208000$
- $208000 \times 1.0816 = 225000$
- $225000 - 10320 = 214680$
- $P \left(1 + \frac{R}{100}\right)^n$
- $P \left(1 + \frac{R}{100}\right)^{2n}$




VIDEO SOLUTION

Question 47

If a seller gives a discount of 15% on retail price, she still makes a profit of 2%. Which of the following ensures that she makes a profit of 20%?

- A Give a discount of 5% on retail price.
- B Give a discount of 2% on retail price.
- C Increase the retail price by 2%.

D Sell at retail price.

cracku

If a seller gives a discount of 15% on retail price, she still makes a profit of 2%. Which of the following ensures that she makes a profit of 20%?

A) Give a discount of 5% on retail price.
B) Give a discount of 2% on retail price.
C) Increase the retail price by 2%.
D) Sell at retail price.

$\text{COST Price} = 100$
 $\text{SELL Price} = 100 \times 0.85 = 85$
 $102 = 0.85 \times \frac{100}{0.85} \times 1.02$
 $102 = 0.85 \times \frac{100}{0.85} \times 1.02$
 $102 = 0.85 \times \frac{100}{0.85} \times 1.02$



VIDEO SOLUTION

Question 48

A person invested a total amount of Rs 15 lakh. A part of it was invested in a fixed deposit earning 6% annual interest, and the remaining amount was invested in two other deposits in the ratio 2 : 1, earning annual interest at the rates of 4% and 3%, respectively. If the total annual interest income is Rs 76000 then the amount (in Rs lakh) invested in the fixed deposit was

cracku

A person invested a total amount of Rs 15 lakh. A part of it was invested in a fixed deposit earning 6% annual interest, and the remaining amount was invested in two other deposits in the ratio 2 : 1, earning annual interest at the rates of 4% and 3%, respectively. If the total annual interest income is Rs 76000 then the amount (in Rs lakh) invested in the fixed deposit was

$15 = F + 2x + x$
 $0.76 = 0.06F + 0.04(2x) + 0.03(x)$
 $0.76 = 0.06F + 0.08(2x) + 0.03(x)$
 $76 = 6F + 16x + 3x$
 $76 = 6F + 19x$



VIDEO SOLUTION

Logical Reasoning for CAT Questions (download pdf)

Instructions

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. Among the third and fourth daughters, he distributed the remaining gold and silver bars equally both together accounting each for 20% of the property remaining after the previous distribution. He also gave the third and fourth daughters Rs. 25000 more than what the second daughter had received in cash.

Question 49

Total property of Ghosh Babu (in Rs.lakh) is

A 5.0

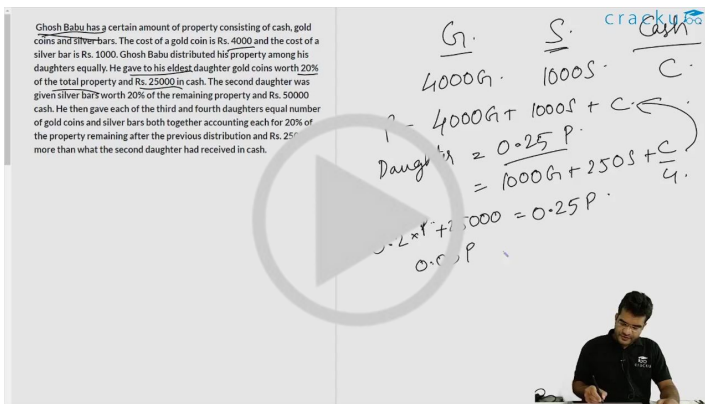
B 7.5

C 10.0

D 12.5.

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. He then gave each of the third and fourth daughters equal number of gold coins and silver bars both together accounting each for 20% of the property remaining after the previous distribution and Rs. 25000 more than what the second daughter had received in cash.

$$P = 4000G + 1000S + C$$
$$\text{Daughter 1} = \frac{0.25P}{1000G + 250S + \frac{C}{4}}$$
$$0.25P + 25000 = 0.25P$$
$$0.25P$$



VIDEO SOLUTION

Instructions

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 66.67%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

Question 50

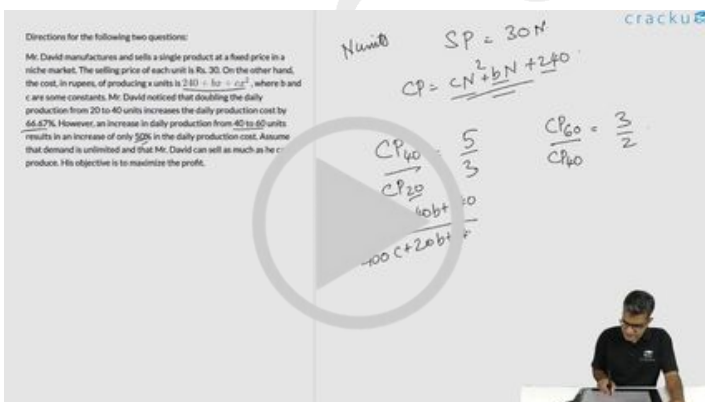
What is the maximum daily profit, in rupees, that Mr. David can realize from his business?

- A 620
- B 920
- C 840
- D 760
- E Cannot be determined

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 66.67%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

$$SP = 30N$$
$$CP = \frac{CN^2 + bN + 240}{2}$$
$$\frac{CP_{60}}{CP_{40}} = \frac{5}{3}$$
$$\frac{CP_{60}}{CP_{40}} = \frac{3}{2}$$
$$40b + 100c + 240b + 100c + 240$$



VIDEO SOLUTION

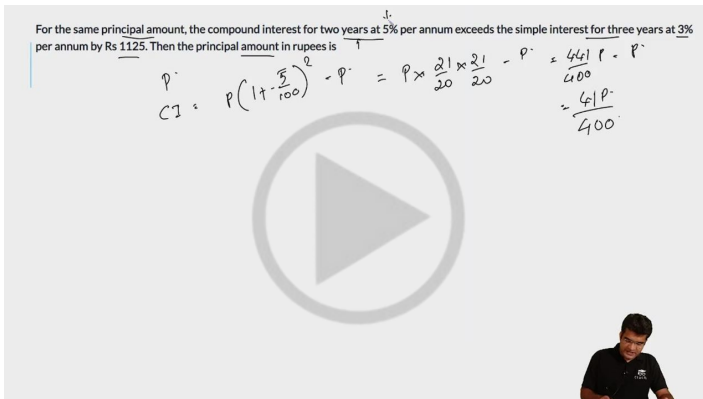
Instructions

For the following questions answer them individually

Question 51

For the same principal amount, the compound interest for two years at 5% per annum exceeds the simple interest for three years at 3% per annum by Rs 1125. Then the principal amount in rupees is

For the same principal amount, the compound interest for two years at 5% per annum exceeds the simple interest for three years at 3% per annum by Rs 1125. Then the principal amount in rupees is

$$CI = P \left(1 + \frac{5}{100}\right)^2 - P = P \times \frac{21}{20} \times \frac{21}{20} - P = \frac{41}{400} P - P = \frac{41P}{400}$$


VIDEO SOLUTION

[Quantitative Aptitude for CAT Questions \(download pdf\)](#)

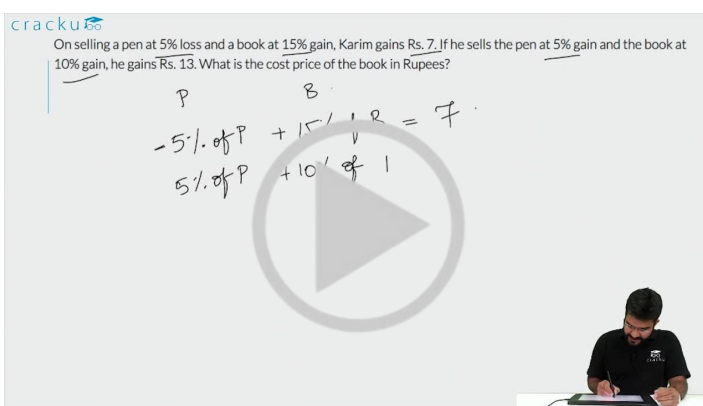
Question 52

On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs. 7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs. 13. What is the cost price of the book in Rupees?

- A 95
- B 85
- C 80
- D 100

cracku

On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs. 7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs. 13. What is the cost price of the book in Rupees?

$$\begin{array}{l} P \quad B \\ -5\% \text{ of } P + 15\% \text{ of } B = 7 \\ 5\% \text{ of } P + 10\% \text{ of } B = 13 \end{array}$$


VIDEO SOLUTION

Question 53

The owner of an art shop conducts his business in the following manner: every once in a while he raises his prices by X%, then a while later he reduces all the new prices by X%. After one such updown cycle, the price of a painting decreased by Rs. 441. After a second up-down cycle the painting was sold for Rs. 1,944.81. What was the original price of the painting?

- A Rs. 2,756.25
- B Rs. 2,256.25
- C Rs. 2,500
- D Rs. 2,000

The owner of an art shop conducts his business in the following manner: every once in a while he raises his prices by X%, then a while later he reduces all the new prices by X%. After one such up-down cycle, the price of a painting decreased by Rs. 441. After a second up-down cycle the painting was sold for Rs. 1,944.81. What was the original price of the painting?

A) Rs. 2,756.25 B) Rs. 2,256.25
C) Rs. 2,500 D) Rs. 2,000

$P - P(1+x)(1-x) = 441$
 $P - P(1-x^2) = 441$
 $Px^2 = 441$ $P(1+x)(1-x)(1+x)(1-x) = 1944.81$

VIDEO SOLUTION

Question 54

Anil invests some money at a fixed rate of interest, compounded annually. If the interests accrued during the second and third year are ₹ 806.25 and ₹ 866.72, respectively, the interest accrued, in INR, during the fourth year is nearest to

- A 929.48
- B 934.65
- C 931.72
- D 926.84

Anil invests some money at a fixed rate of interest, compounded annually. If the interests accrued during the second and third year are ₹ 806.25 and ₹ 866.72, respectively, the interest accrued, in INR, during the fourth year is nearest to

A) 929.48 B) 934.65
C) 931.72 D) 926.84

$P(1+x)^2 - P(1+x) = 806.25$
 $P(1+x) = 806.25$
 $P(1+x)^3 - P(1+x)^2 = 866.72$
 $P(1+x)^2 = 866.72$
 $P(1+x)^4 - P(1+x)^3 = ?$

VIDEO SOLUTION

Know the CAT Percentile Required for IIM Calls

Question 55

Bank A offers 6% interest rate per annum compounded half-yearly. Bank B and Bank C offer simple interest but the annual interest rate offered by Bank C is twice that of Bank B. Raju invests a certain amount in Bank B for a certain period and Rupa invests ₹ 10,000 in Bank C for twice that period. The interest that would accrue to Raju during that period is equal to the interest that would have accrued had he invested the same amount in Bank A for one year. The interest accrued, in INR, to Rupa is

- A 3436
- B 2436
- C 2346
- D 1436

Bank A offers 6% interest rate per annum compounded half-yearly, Bank B and Bank C offer simple interest but the annual interest rate offered by Bank C is twice that of Bank B. Raju invests a certain amount in Bank B for a certain period and Rupa invests ₹ 10,000 in Bank C for twice that period. The interest that would accrue to Raju during that period is equal to the interest that would have accrued had he invested the same amount in Bank A for one year. The interest accrued, in INR, to Rupa is

A) 3436
C) 2346

3% every 6 months
Raju
1 yr
Rupa
2 yr
10,000
P(1.03)²

VIDEO SOLUTION

Question 56

A person invested a certain amount of money at 10% annual interest, compounded half-yearly. After one and a half years, the interest and principal together became Rs.18522. The amount, in rupees, that the person had invested is

A person invested a certain amount of money at 10% annual interest, compounded half-yearly. After one and a half years, the interest and principal together became Rs.18522. The amount, in rupees, that the person had invested is

I = 10% p.a. = 5%
T = 1.5 yr. = 3 yr

R

VIDEO SOLUTION

Instructions

Answer the questions based on the following information. A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000, which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100.

Question 57

If he is able to sell only 1,200 out of 1,500 watches he has made in the season (and the rest 300 are sold out of season), then he has made a profit of

- A Rs. 90,000
- B Rs. 75,000
- C Rs. 45,000
- D Rs. 60,000

Answer the questions based on the following information. A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000 which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100.

If he is able to sell only 1,200 out of 1,500 watches he has made in the season (and the rest 300 are sold out of season), then he has made a profit of

A) Rs. 90,000
B) Rs. 75,000
C) Rs. 45,000
D) Rs. 60,000

$TCP = 150 \times 1500 + 30000 = 225000 + 30000 = 255000$
 $TSP = 1200 \times 250 + 300 \times 100 = 300000 + 30000 = 330000$
 Profit = $TSP - TCP = 330000 - 255000 = 75000$

VIDEO SOLUTION

Join MBA Telegram Group

Instructions

For the following questions answer them individually

Question 58

The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

- A 1500
- B 2000
- C 2500
- D 3000

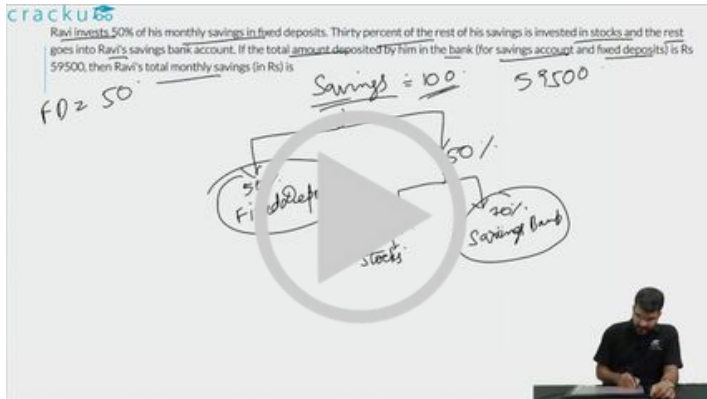
The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

$M \xrightarrow{1.1C} W \xrightarrow{1.3C} R \xrightarrow{1.5C} C$

$4290 = 1.5C \Rightarrow C = \frac{4290}{1.5} = 2860$
 $2860 = 1.3C \Rightarrow C = \frac{2860}{1.3} = 2200$
 $2200 = 1.1C \Rightarrow C = \frac{2200}{1.1} = 2000$

Question 59

Ravi invests 50% of his monthly savings in fixed deposits. Thirty percent of the rest of his savings is invested in stocks and the rest goes into Ravi's savings bank account. If the total amount deposited by him in the bank (for savings account and fixed deposits) is Rs 59500, then Ravi's total monthly savings (in Rs) is



Question 60

The price of a Maruti car rises by 30% while the sales of the car come down by 20%. What is the percentage change in the total revenue?

- A -4%
- B -2%
- C +4%
- D +2%

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Question 61

Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu at 20% profit. Then, Kanu sells it back to Minu at 20% loss. Finally, Minu sells the same pair of sunglasses to Tanu. If the total profit made by Minu from all her transactions is Rs.500, then the percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is

- A 35.42%
- B 52%
- C 31.25%
- D 26%

cracku

Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu at 20% profit. Then, Kanu sells it back to Minu at 20% loss. Finally, Minu sells the same pair of sunglasses to Tanu. If the total profit made by Minu from all her transactions is Rs.500, then the percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is

A) 35.42%
B) 52%
C) 31.25%

Handwritten notes:
Minu
CP = Rs. 1000
SP = Rs. 1200
20%
Kanu
CP = Rs. 1200
SP = 0.8 × 1200 = Rs. 960

VIDEO SOLUTION

Instructions

Directions for the following two questions: Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of - 3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of - 2.5%, while a fall will entail a return of + 2%.

Question 62

What strategy will maximize the guaranteed return to Shabnam?

- A 100% in option A
- B 36% in option B and 64% in option C
- C 64% in option B and 36% in option C
- D 1/3 in each of the three options
- E 30% in option A, 32% in option B and 38% in option C

cracku

Directions for the following two questions: Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of - 3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of - 2.5%, while a fall will entail a return of + 2%.

Question (1)
The maximum guaranteed return to Shabnam is

A) 0.25%
B) 0.50%
C) 0.20%
D) 0.15%
E) 0.30%

Handwritten notes:
Option A
Option B
Option C
100-A

VIDEO SOLUTION

Instructions

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. Among the third and fourth daughters, he distributed the remaining gold and silver bars equally both together accounting each for 20% of the property remaining after the previous distribution. He also gave the third and fourth daughters Rs. 25000 more than what the second daughter had received in cash.

Question 63

If the ratio of the gold to silver bars that Ghosh Babu had is 7:27, the number of silver bars he has is

- A 90
- B 60
- C 75
- D 135

VIDEO SOLUTION

Cracku CAT Success Stories

Question 64

The amount of property in gold and silver possessed by Ghosh Babu is

- A 2,25,000
- B 2,75,000
- C Rs. 4,25,000
- D None of these

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. He then gave each of the third and fourth daughters equal number of gold coins and silver bars both together accounting each for 20% of the property remaining after the previous distribution and Rs. 25000 more than what the second daughter had received in cash.

Handwritten notes on a whiteboard:

$$G: 4000G, S: 1000S, C: \text{Cash}$$

$$P = 4000G + 1000S + C$$

$$\text{Daughter 1} = 0.25P = 1000G + 250S + \frac{C}{4}$$

$$0.2 \times P + 25000 = 0.25P$$

$$0.05P = 25000$$

VIDEO SOLUTION

Instructions

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakhs to buy 100 pieces. He can purchase a colour TV at Rs.10,000 and a VCR at Rs.15,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,500. His objective is to maximize profits. Assume that he can sell all the items that he stocks.

Question 65

If the dealer would have managed to get an additional space to stock 20 more items, then for maximizing profit, the ratio of number of VCRs and number of TVs that he should stock is

- A 7 : 3
- B 0
- C 1 : 2
- D None of these

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakhs to buy 100 pieces. He can purchase a colour TV at Rs.10,000 and a VCR at Rs.15,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,500. His objective is to maximize profits. Assume that he can sell all the items that he stocks. If the dealer would have managed to get an additional space to stock 20 more items, then for maximizing profit, the ratio of number of VCRs and number of TVs that he should stock is

Handwritten notes on a whiteboard:

$$a + b = 120$$

$$10K + 15K = 12K + 17.5K$$

$$a = 70, b = 50$$

$$a:b = 7:5$$

VIDEO SOLUTION

Instructions

For the following questions answer them individually

Question 66

Gopal went to a fruit market with certain amount of money. With this money he can buy either 50 oranges or 40 mangoes. He retains 10% of the money for taxi fare. If he buys 20 mangoes, then the number of oranges he can buy is

- A 25

- B 18
- C 20
- D None of these

cracku

Gopal went to a fruit market with certain amount of money. With this money he can buy either 50 oranges or 40 mangoes. He retains 10% of the money for taxi fare. If he buys 20 oranges, then the number of mangoes he can buy is

A) 25
B) 18
C) 20
D) None of these

50 oranges = 40 mangoes = Rs. 200.
1 orange = Rs. 4.
20 =

VIDEO SOLUTION

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Question 67

A man earns $x\%$ on the first Rs. 2,000 and $y\%$ on the rest of his income. If he earns Rs. 700 from income of Rs. 4,000 and Rs. 900 from Rs. 5,000 of income, find $x\%$.

- A 20%
- B 15%
- C 25%
- D None of these

cracku

A man earns $x\%$ on the first Rs. 2,000 and $y\%$ on the rest of his income. If he earns Rs. 700 from income of Rs. 4,000 and Rs. 900 from Rs. 5,000 of income, find $x\%$.

A) 20%
B) 15%
C) 25%
D) None of these

2000 + 2000 = 4000
 $x\%$ $y\%$ = 700
2000 + 3000 = 5000
 $x\%$ $y\%$ = 900
 $x = ?$

VIDEO SOLUTION

Question 68

A dealer buys dry fruits at Rs. 100, Rs. 80 and Rs. 60 per kilogram. He mixes them in the ratio 3 : 4 : 5 by weight, and sells at a profit of 50%. At what price per kilogram does he sell the dry fruits?

- A Rs. 80

- B Rs. 100
- C Rs. 95
- D None of these

A dealer buys dry fruits at Rs. 100, Rs. 80 and Rs. 60 per kilogram. He mixes them in the ratio 3 : 4 : 5 by weight, and sells at a profit of 50%. At what price per kilogram does he sell the dry fruits?

A) Rs. 80
B) Rs. 100
C) Rs. 95
D) None of these

VIDEO SOLUTION

Question 69

After allowing a discount of 11.11%, a trader still makes a gain of 14.28%. At how many percentage above the cost price does he mark on his goods?

- A 28.56%
- B 35%
- C 22.22%
- D None of these

After allowing a discount of 11.11%, a trader still makes a gain of 14.28%. At how many percentage above the cost price does he mark on his goods?

A) 28.56%
B) 35%
C) 22.22%
D) None of these

VIDEO SOLUTION

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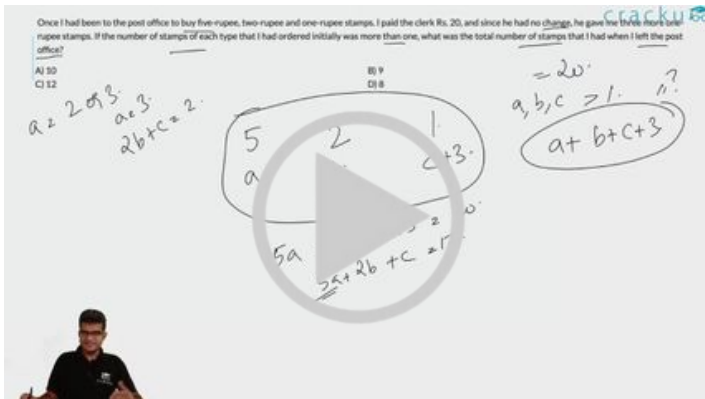
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Question 70

Once I had been to the post office to buy five-rupee, two-rupee and one-rupee stamps. I paid the clerk Rs. 20, and since he had no change, he gave me three more one-rupee stamps. If the number of stamps of each type that I had ordered initially was more than one, what was the total number of stamps that I had when I left the post office?

- A 10
- B 9
- C 12
- D 8

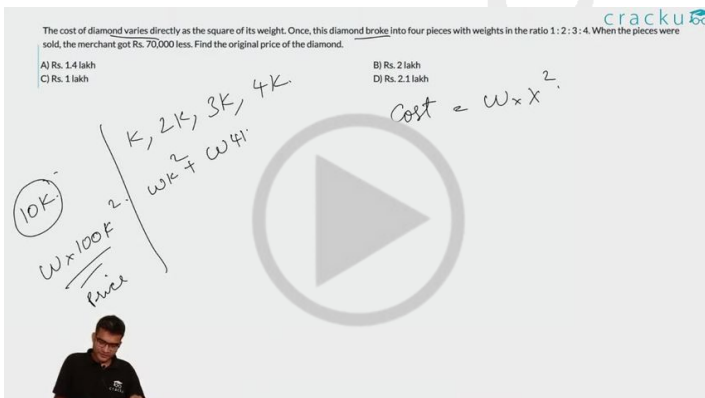


VIDEO SOLUTION

Question 71

The cost of diamond varies directly as the square of its weight. Once, this diamond broke into four pieces with weights in the ratio 1 : 2 : 3 : 4. When the pieces were sold, the merchant got Rs. 70,000 less. Find the original price of the diamond.

- A Rs. 1.4 lakh
- B Rs. 2 lakh
- C Rs. 1 lakh
- D Rs. 2.1 lakh



VIDEO SOLUTION

Question 72

John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is

cracku

John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is

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Question 73

A merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him. He sells the same cloth at a rate of Rs.110 per meter but cheats his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers. If the merchant provides a 5% discount, the resulting profit earned by him is

- A 4.2%
- B 9.7%
- C 15.5%
- D 16%

cracku

A merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him. He sells the same cloth at a rate of Rs.110 per meter but cheats his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers. If the merchant provides a 5% discount, the resulting profit earned by him is

A) 4.2%
C) 15.5%

Rs. 100
1m
0.05
Rs. 110/m → 0.95

Rs. 9.7%

VIDEO SOLUTION

Instructions

Answer the questions based on the following information. A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000, which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100.

Question 74

If he produces 1,500 watches, what is the number of watches that he must sell during the season in order to break-even, given that he is able to sell all the watches produced?

- A 500
- B 700
- C 800
- D 1,000

Answer the questions based on the following information. A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000 which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100. If he is able to sell only 1,200 out of 1,500 watches he has made in the season (and the rest 300 are sold out of season), then he has made a profit of

A) Rs. 90,000
B) Rs. 100,000
C) Rs. 45,000
D) Rs. 60,000

$TCP = 1500 \times 150 + 30000 = 225000 + 30000 = 255000$
 $TSP = 1200 \times 250 + 300 \times 100 = 300000 + 30000 = 330000$
 $Profit = TSP - TCP = 330000 - 255000 = 75000$

[VIDEO SOLUTION](#)

Instructions

For the following questions answer them individually

Question 75

Alex invested his savings in two parts. The simple interest earned on the first part at 15% per annum for 4 years is the same as the simple interest earned on the second part at 12% per annum for 3 years. Then, the percentage of his savings invested in the first part is

- A 37.5%
- B 62.5%
- C 60%
- D 40%

cracku

Alex invested his savings in two parts. The simple interest earned on the first part at 15% per annum for 4 years is the same as the simple interest earned on the second part at 12% per annum for 3 years. Then, the percentage of his savings invested in the first part is

A) 37.5%
B) 62.5%
C) 60%
D) 40%

Handwritten solution:

$$\frac{A \times 15 \times 4}{100} = \frac{B \times 12 \times 3}{100}$$

Video solution showing a person writing the solution.

VIDEO SOLUTION

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Instructions

Answer the following questions based on the following information.

A company purchases components A and B from Germany and USA respectively. A and B form 30% and 50% of the total production cost. Current gain is 20%. Due to change in the international scenario, cost of the German mark increased by 30% and that of USA dollar increased by 22%. Due to market conditions, the selling price cannot be increased by more than 10%.

Question 76

If the USA dollar becomes cheap by 12% over its original cost and the cost of German mark increased by 20%, what will be the gain? (The selling price is not altered.)

- A 10%
- B 20%
- C 15%
- D 7.5%

VIEW SOLUTION

Instructions

For the following questions answer them individually

Question 77

Aman invests Rs 4000 in a bank at a certain rate of interest, compounded annually. If the ratio of the value of the investment after 3 years to the value of the investment after 5 years is 25 : 36, then the minimum number of years required for the value of the investment to exceed Rs 20000 is

cracku

Aman invests Rs 4000 in a bank at a certain rate of interest, compounded annually. If the ratio of the value of the investment after 3 years to the value of the investment after 5 years is 25 : 36, then the minimum number of years required for the value of the investment to exceed Rs 20000 is _____

Answer: _____

$$\frac{4000 \times (1+R)^3}{4000 \times (1+R)^5} = \frac{25}{36}$$

$$(1+R)^2 = 1.44$$

$$R = 20\%$$

$$4000(1.2)^N > 20000$$

(1)



VIDEO SOLUTION



Answers

1.A	2.B	3.A	4.C	5.D	6.A	7.D	8.20920
9.A	10.C	11.B	12.20808	13.1000	14.4000	15.A	16.B
17.B	18.A	19.B	20.C	21.20000	22.D	23.12	24.D
25.C	26.D	27.C	28.B	29.A	30.D	31.A	32.A
33.A	34.160	35.C	36.B	37.D	38.2160	39.D	40.A
41.C	42.407	43.A	44.C	45.20	46.B	47.D	48.9
49.A	50.D	51.90000	52.C	53.A	54.C	55.B	56.16000
57.B	58.B	59.70000	60.C	61.C	62.B	63.D	64.B
65.B	66.C	67.B	68.D	69.A	70.A	71.C	72.121000
73.C	74.B	75.A	76.B	77.9			

Explanations

1. A

Let the total production cost be 100.

Hence, selling price is 120.

Price of German component A is 30 and the price of the US component B is 50

After change in exchange rate, price of German component is $30 \times 1.3 = 39$
and price of US component is $50 \times 1.22 = 61$

Total increase equals $39 + 61 - 30 - 50 = 20$

Hence, the minimum production cost is $100 + 20 = 120$

The maximum possible selling price is $120 \times 110\% = 132$.

So, maximum possible gain is $(132 - 120) / 120 = 10\%$

[VIEW SOLUTION](#)

2. B

After the first game, Ghosh Babu picked 8 of clubs => He gets Rs 8. Then the dealer picked Queen of clubs => Ghosh Babu pays Rs 16 => Ghosh Babu is at a loss of Rs 8 after 1st game.

After the second game, Ghosh Babu picked 10 of hearts => He gets Rs 10. Then the dealer picked 2 of spades => Ghosh Babu gets another Rs 10 => Ghosh Babu is now at a profit of Rs 12.

After the third game, Ghosh Babu picked six of diamonds => He gets Rs 6. Then the dealer picked ace of hearts => Ghosh Babu pays Rs 6 to dealer => Ghosh Babu is still at a profit of Rs 12.

In the fourth game, Ghosh Babu picks 8 of spades => He gets Rs 8. The the dealer picks jack of spades => Ghosh Babu pays Rs 16 to dealer => Ghosh Babu is at a profit of Rs 4.

Hence, the maximum profit earned is Rs 12.

The maximum loss that Ghosh Babu had was Rs 8.

He must have had at least Rs 8 so that he did not have to borrow any amount from others.

[VIEW SOLUTION](#)

3. A

Let's say the cost of the cloth is x rs per metre. Because of the faulty meter, he is paying x for 120 cms when buying.

So cost of 100 cms = $100x / 120$.

He is selling 80 cms for x, so selling price of 100cms of cloth is $100x/80$.

discount = 20%

so the effective selling price is $.8 \cdot 100x/80 = x$

profit = SP-CP = $x - 100x/120 = x/6$

Profit % = $x/6$ divided by $100x/120 = 20\%$

 VIDEO SOLUTION

4. C

Let a, b and c be the percentages of amount invested in options A, B and C respectively $\Rightarrow a + b + c = 100$

Return attained if there is a rise in the stock market $\Rightarrow 0.001a + 0.05b - 0.025c$

Return attained if there is a fall in the stock market $\Rightarrow 0.001a - 0.03b + 0.02c$

Maximum guaranteed return is attained when both are equal because it is indifferent to rise and fall in the market.

$$0.001a + 0.05b - 0.025c = 0.001a - 0.03b + 0.02c$$

$$\Rightarrow 0.08b = 0.045c \Rightarrow 16b = 9c$$

Let's put the values for a, b and c that satisfy the above equation.

$$b = 9, c = 16, a = 75 \Rightarrow \text{return} = 0.125$$

$$b = 18, c = 32, a = 50 \Rightarrow \text{return} = 0.15$$

$$b = 27, c = 48, a = 25 \Rightarrow \text{return} = 0.175$$

$$b = 36, c = 64, a = 0 \Rightarrow \text{return} = 0.2$$

Hence, the maximum guaranteed return is 0.2%

 VIDEO SOLUTION

5. D

After the first game, Ghosh Babu picked 8 of clubs $\Rightarrow$ He gets Rs 8. Then the dealer picked Queen of clubs $\Rightarrow$ Ghosh Babu pays Rs 16 $\Rightarrow$ Ghosh Babu is at a loss of Rs 8 after 1st game.

After the second game, Ghosh Babu picked 10 of hearts $\Rightarrow$ He gets Rs 10. Then the dealer picked 2 of spades $\Rightarrow$ Ghosh Babu gets another Rs 10 $\Rightarrow$ Ghosh Babu is now at a profit of Rs 12.

After the third game, Ghosh Babu picked six of diamonds $\Rightarrow$ He gets Rs 6. Then the dealer picked ace of hearts $\Rightarrow$ Ghosh Babu pays Rs 6 to dealer $\Rightarrow$ Ghosh Babu is still at a profit of Rs 12.

In the fourth game, Ghosh Babu picks 8 of spades $\Rightarrow$ He gets Rs 8. The the dealer picks jack of spades $\Rightarrow$ Ghosh Babu pays Rs 16 to dealer $\Rightarrow$ Ghosh Babu is at a profit of Rs 4.

As Ghosh Babu earned a profit of Rs 4 and now he has Rs 100, he initially would have had $100 - 4 = \text{Rs } 96$.

 VIEW SOLUTION

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6. A

After the first game, Ghosh Babu picked 8 of clubs $\Rightarrow$ He gets Rs 8. Then the dealer picked Queen of clubs $\Rightarrow$ Ghosh Babu pays Rs 16 $\Rightarrow$ Ghosh Babu is at a loss of Rs 8 after 1st game.

After the second game, Ghosh Babu picked 10 of hearts $\Rightarrow$ He gets Rs 10. Then the dealer picked 2 of spades $\Rightarrow$ Ghosh Babu gets another Rs 10 $\Rightarrow$ Ghosh Babu is now at a profit of Rs 12.

After the third game, Ghosh Babu picked six of diamonds => He gets Rs 6. Then the dealer picked ace of hearts => Ghosh Babu pays Rs 6 to dealer => Ghosh Babu is still at a profit of Rs 12.

In the fourth game, Ghosh Babu picks 8 of spades => He gets Rs 8. The the dealer picks jack of spades => Ghosh Babu pays Rs 16 to dealer => Ghosh Babu is at a profit of Rs 4.

Hence, the maximum profit earned is Rs 12.

 VIEW SOLUTION

7. **D**

Let the cost price be C and the marked price be M.

Given,

$$0.6 M = 1.2 C$$

$$M = 2C$$

$$\text{CP of 60 toys} = 60C$$

Now only 50 are remaining.

Hence,

$$M(1 - d) * 50 = 72C$$

$$1 - d = 0.72$$

$$d = .28$$

Hence 28%

 VIDEO SOLUTION

8. **20920**

$$\text{The amount with Amal at the end of 1 year} = 12000 * 1.08 + 10000 * 1.03 * 1.03 = 23569$$

$$\text{Interest received by Amal} = 23569 - 22000 = 1569$$

$$\text{Let the amount invested by Bimal} = 100b$$

$$\text{Interest received by Bimal} = 100b * 7.5 * 1 / 100 = 7.5b$$

It is given that the amount of interest received by both of them is the same

$$7.5b = 1569$$

$$b = 209.2$$

$$\text{Amount invested by Bimal} = 100b = 20920$$

 VIDEO SOLUTION

9. **A**

CP of the table at which the shopkeeper procured each table = p

It is given that shopkeeper sold the tables to Amal and Asim at a profit of 20% and at a loss of 20%, respectively

The selling price of the tables = 1.2p and 0.8p to Amal and Asim respectively.

Amal sells his table to Bimal at a profit of 30%

$$\text{So, CP of the table by Bimal (x)} = 1.2p * 1.3 = 1.56p$$

Asim sells his table to Barun at a loss of 30%

$$\text{So, CP of the table by Barun (y)} = 0.7 * 0.8p = 0.56p$$

$$(x - y) / p = (1.56p - 0.56p) / p = p / p = 1$$

 VIDEO SOLUTION

10. C

Let the cost of each bicycle = 100b

CP of 10 bicycles = 1000b

It is given that he sold six of these at a profit of 25% and the remaining four at a loss of 25%

SP of 10 bicycles = $125b \times 6 + 75b \times 4$

= 1050b

Profit = 1050b - 1000b = 50b

50b = 2000

CP = 100b = 4000

 VIDEO SOLUTION



11. B

It is given,

$$\frac{5.5 \times 1 \times 8000}{100} + \frac{5.6 \times 1 \times 5000}{100} + \frac{x \times 1 \times 7000}{100} = \frac{5}{100} \times 20000$$

$$440 + 280 + 70x = 1000$$

$$x = 4\%$$

$$\text{Interest} = \frac{20000 \times 4 \times 1}{100} = \text{Rs } 800$$

The answer is option B.

 VIDEO SOLUTION

12. 20808

Anil invested 22000 for 6 years at 4% interest compounded half-yearly

$$\Rightarrow \text{Amount} = 22000 (1.02)^{12}$$

Let Sunil invest 'S' rupees for 5 years at 4% C.I. half-yearly and 10% S.I. for 1 additional year

$$\Rightarrow \text{Amount} = S (1.02)^{10} (1.1)$$

Given that the both amounts are equal

$$\Rightarrow 22000 (1.02)^{12} = S (1.02)^{10} (1.1)$$

$$\Rightarrow S = \frac{22000 (1.02)^2}{1.1} = 20808$$

 VIDEO SOLUTION

13. 1000

Let the number of pens purchased be n. Then the cost price is 8n. The total expenses incurred would be 8n+W, where W refers to the wage.

$$\text{Then SP in the first case} = 12 \times 100 + 11 \times (n - 100)$$

$$\text{Given profit is 300 in this case: } 1200 + 11n - 1100 - 8n - W = 300 \Rightarrow 3n - W = 200$$

In second case: $1200 + 9n - 900 - 8n - W = -300$ (Loss). $\Rightarrow W - n = 600$.

Adding the two equations: $2n = 800$

$n = 400$.

Thus $W = 600 + 400 = 1000$

 VIDEO SOLUTION

14. 4000

Amount of interest paid by Ishan to Gopal if the borrowed amount is Rs. $(X+Y) = \frac{10}{100} * (X + Y) = 0.1(X+Y)$

Gopal also borrowed Rs. X from Ankit at 8% per annum. Therefore, he has to return Ankit Rs. $0.08X$ as the interest amount on borrowed sum.

Hence, the interest retained by gopal $= 0.1(X+Y) - 0.08X = 0.02X + 0.1Y \dots (1)$

It is given that the net interest retained by Gopal is the same as that accrued to Ankit.

Therefore, $0.08X = 0.02X + 0.1Y$

$\Rightarrow X = (5/3)Y \dots (2)$

Amount of interest paid by Ishan to Gopal if the borrowed amount is Rs. $(X+2Y) = \frac{10}{100} * (X + 2Y) = 0.1X + 0.2Y$

In this case the amount of interest retained by Gopal $= 0.1X + 0.2Y - 0.08X = 0.02X + 0.2Y \dots (3)$

It is given that the interest retained by Gopal increased by Rs. 150 in the second case.

$\Rightarrow (0.02X + 0.2Y) - (0.02X + 0.1Y) = 150$

$\Rightarrow Y = \text{Rs. } 1500$

By substituting value of Y in equation (2), we can say that $X = \text{Rs. } 2500$

Therefore, $(X+Y) = \text{Rs. } 4000$.

 VIDEO SOLUTION

15. A

Assuming the income of Bimla $= 100a$, then the income of Amala will be $120a$.

And the income of Kamala will be $120a * 100/80 = 150a$

If Kamala's income goes down by 4%, then new income of Kamala $= 150a - 150a(4/100) = 150a - 6a = 144a$

If Bimla's income goes up by 10 percent, her new income will be $100a + 100a(10/100) = 110a$

$\Rightarrow$ Hence the Kamala income will exceed Bimla income by $(144a - 110a) * 100/110a = 31$

 VIDEO SOLUTION

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16. B

Selling price of first watch $= 300$

Profit $= 10\%$

cost price $= \frac{300}{1.1}$

Selling price of second watch $= 300$

Loss $= 10\%$

cost price $= \frac{300}{0.9}$

Total selling price of transaction $= 600$

$$\text{Total cost price of transaction} = 300\left(\frac{10}{11} + \frac{10}{9}\right) = 600\left(\frac{100}{99}\right)$$

$$\text{Loss} = 600\left(\frac{100}{99} - 1\right)$$

$$\% \text{loss} = \left(600\left(\frac{100}{99} - 1\right)\right) \div \left(600\left(\frac{100}{99}\right)\right) \times 100 = 1$$

 VIDEO SOLUTION

17. B

Let the cost of good mangoes be $2x$ per kg. The cost of medium mangoes be x per kg.

$$\text{CP of good mangoes} = 160x$$

$$\text{CP of medium mangoes} = 40x$$

$$\text{His selling price} = 0.9 \times 2x = 1.80x$$

$$\text{Therefore, total revenue generated by selling all the mangoes} = 120 \times 1.8x = 216x$$

$$\text{Hence, the profit \%} = \frac{16x}{200x} \times 100 = 8\%$$

 VIDEO SOLUTION

18. A

Let the CP of the each toy be " x ". CP of 12 toys will be " $12x$ ". Now the shopkeeper made a 10% profit on CP. This means that

$$12x(1.1) = 2112 \text{ or } x = 160. \text{ Hence the CP of each toy is ₹160.}$$

Now let the SP of each toy be " m ". Now he sold 8 toys at 20% discount. This means that $8m(0.8)$ or $6.4m$

He sold 4 toys at an additional 25% discount. $4m(0.8)(0.75) = 2.4m$ Now $6.4m + 2.4m = 8.8m = 2112$ or $m = 240$

Hence CP = 160 and SP = 240. Hence profit percentage is 50%.

 VIDEO SOLUTION

19. B

$$\text{Cost of 20 units} = 240 + 20b + 400c$$

$$\text{Cost of 40 units} = 240 + 40b + 1600c = \frac{5}{3} * (240 + 20b + 400c) \Rightarrow 720 + 120b + 4800c = 1200 + 100b + 2000c$$

$$\Rightarrow 480 = 20b + 2800c \Rightarrow 120 = 5b + 700c$$

$$\text{Cost of 60 units} = 240 + 60b + 3600c = \frac{3}{2} (240 + 40b + 1600c) \Rightarrow 480 + 120b + 7200c = 720 + 120b + 4800c$$

$$\Rightarrow 240 = 2400c \Rightarrow c = 1/10 \text{ and } b = 10$$

Let the number of items needed for max profit be k

$$\text{CP} = 240 + 10k + \frac{k^2}{10}$$

$$\text{SP} = 30k$$

$$\text{Profit} = \text{SP} - \text{CP} = 30k - 240 - 10k - \frac{k^2}{10} = 20k - 240 - \frac{k^2}{10}$$

$$\text{or Profit} = \frac{1}{10}(-k^2 + 200k - 2400)$$

$$\text{or, Profit} = \frac{1}{10}(-(k^2 - 200k + 2400))$$

$$\text{or, Profit} = \frac{1}{10}(-(k^2 - 200k + 2400 + 7600 - 7600))$$

$$\text{or, Profit} = \frac{1}{10}(-(k^2 - 200k + 10000) + 7600)$$

$$\text{or, Profit} = \frac{1}{10}(-(k - 100)^2 + 7600)$$

To maximise the value of Profit, $-(k - 100)^2$ must be 0.

So, k must be equal to 100.

Hence, option B is the correct answer.

 VIDEO SOLUTION

20. C

Assuming the maximum marks = 100a, then Meena got 40a

After increasing her score by 50%, she will get $40a(1+50/100)=60a$

Passing score = $60a+35$

Post review score after 20% increase = $60a*1.2=72a$

=>Hence, $60a+35=72a$

=> $12a=42 \Rightarrow a=3.5$

=> maximum marks = 350 and passing marks = $210+35=245$

=> Passing percentage = $245*100/350 = 70$

 VIDEO SOLUTION



21. 20000

Let the price of desktop and laptop be x,y respectively.

Given,

$x+y=50000$...(i)

$12.x+0.9y=50000(1.02)=51000$...(ii)

(ii)-0.9(i) gives

$0.3x=6000 \Rightarrow x=20000$.

 VIDEO SOLUTION

22. D

Let 'x' be the strength of group G. Based on the information, $0.65x$ constitutes of literate people {the rest $0.35x$ = illiterate}

Of this $0.65x$, 75% are old people $= (0.75)0.65x$ old literates. The total number of old people in group G is $0.72x$ {72% of the total}. Thus, the total number of old people who are illiterate = $0.72x - 0.4875x = 0.2325x$.

This is $\frac{0.2325x}{0.35x} \times 100 \approx 66\%$ of the total number of illiterates. Hence, Option D is the correct answer.

 VIDEO SOLUTION

23. 12

Let their individual Amounts be equal after 't' years. Let their initial investments amount to A_V and A_J ;

$A_V = 10,000 \left(1 + \frac{5t}{100}\right)$ and $A_J = 8,000 \left(1 + \frac{10(t-2)}{100}\right)$

Equating both: $10,000 \left(1 + \frac{5t}{100}\right) = 8,000 \left(1 + \frac{10(t-2)}{100}\right)$

On simplifying both sides, we get: $15t = 180$; $t = 12$

VIDEO SOLUTION

24. D

Let the total marks be $100x$

Marks obtained by Bishnu = $52x$

Marks obtained by Asha = $64x$

Marks obtained by Ramesh = $52x+23$

Marks obtained by Ramesh = $64x-34$

$$\Rightarrow 52x+23 = 64x-34$$

$$\Rightarrow x = \frac{19}{4}$$

Marks obtained by Geeta = $84x = 84 \cdot \frac{19}{4} = 399$

VIDEO SOLUTION

25. C

Let the cost price of 1kg of sugar = Rs 100

The total cost price of 35 kg = Rs 3500

Marked up price per kg = Rs 120

Given, the final profit is 15% $\Rightarrow$ Final SP of 35 kg = $3500 \cdot 1.15 = \text{Rs } 4025$

First 5 kg's are sold at 20% marked up price $\Rightarrow SP_1 = 5 \cdot 100 \cdot 1.2 = \text{Rs } 600$

Next 15 kgs are sold after giving 10% discount $\Rightarrow SP_2 = 15 \cdot 100 \cdot 1.2 \cdot 0.9 = 1620$

3kgs of sugar got wasted

$\Rightarrow$ 23 kg of sugar was sold at Rs $(600 + 1620) = \text{Rs } 2220$

Remaining 12kg should be sold at Rs $4025 - 2220 = \text{Rs } 1805$

$\Rightarrow$ SP of 1kg = $1805/12 \simeq 150$

Hence, the seller should further mark up by $\frac{(150-120)}{120} \cdot 100 = 25\%$

VIDEO SOLUTION

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26. D

In total Raju bets 6000Rs and ends up with no profit - no loss. So there are 3 possibilities.

- 1) White comes 2nd, Black comes 4th and Red comes 5th.
- 2) Black comes 1st, White comes 3rd and Red comes 4th or 5th.
- 3) Black comes 2nd, Red comes 3rd and White comes 4th or 5th.

So there can never be 3 horses between white and red according to above to possibilities. Hence option D cannot be true.

VIDEO SOLUTION

27. C

There are total 3 cases which satisfies the condition "no profit and no loss."

Case 1: White comes 2nd.(remaining two horses(red/black) come 4th/5th)

Profit from white horse = Final Amount - Initial Amount = $2000 \cdot 3 - 2000 = 4000$

Loss from Red and Black horse = $3000 + 1000 = 4000$

Net profit = $4000 - 4000 = 0$

Case 2: Black, Red come second, third respectively. (remaining one horse (white) comes 4th/5th)

Profit from Black = $1000 \times 3 - 1000 = 2000$

Profit from Red = $3000 - 3000 = 0$

Loss from white = 2000

Net profit = $2000 - 2000 = 0$

Case 3: black, white come first, third respectively. (remaining one horse (red) comes 4th/5th)

Profit from Black = $1000 \times 4 - 1000 = 3000$

Profit from White = $2000 - 2000 = 0$

Loss from Red = 3000

Net Profit = $3000 - 3000 = 0$

And it is mentioned that grey case 4th. $\Rightarrow$ case 1 is wrong. (because, in that case red, black should come 4th, 5th)

So option C cannot be true.

 VIDEO SOLUTION

28. B

As we know, formulae of compound interest for 2 years will be:

$$P\left(1 + \frac{r}{100}\right)^2 = 625 \quad (\text{Where } r \text{ is rate, } P \text{ is principal amount})$$

For 3 years:

$$P\left(1 + \frac{r}{100}\right)^3 = 675$$

Dividing above two equations we will get $r = 8\%$

 VIDEO SOLUTION

29. A

Let's say price of turban is x .

So total price for 12 months will be = $90 + x$

$$\text{total price for 9 months} = \frac{(90+x) \times 9}{12} = (65 + x)$$

By solving above equation, we will get value of $x = 10$.

 VIDEO SOLUTION

30. D

Raj invested Rs 10000 in the first year. Assuming the loss he faced was $x\%$.

The amount after 1 year is $10,000 \times (1 - x/100) = 10000 - 100x$.

Given the balance was greater than Rs 5000 and hence $x < 50$ percent.

When Raj invested this amount in the second year he earned a profit which is five times that of the first-year percentage.

Hence the amount after the second year is : $(10000 - 100x)\left(1 + \frac{5x}{100}\right)$.

Raj gained a total of 35 percent over the period of two years and hence the 35 percent is Rs 3500.

Hence the final amount is Rs 13,500.

$$(10000 - 100x)(1 + \frac{5 \cdot x}{100}) = 13,500$$

$$(100 + 5 \cdot x) \cdot (100 - x) = 13500$$

$$10000 - 100 \cdot x + 500 \cdot x - 5 \cdot x^2 = 13500.$$

$$5x^2 - 400x + 3500 = 0$$

Solving the equation the roots are :

$$x = 10, x = 70.$$

Since $x < 50$, $x = 10$ percent.

 VIDEO SOLUTION



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31. A

Let the amount invested by Anil Bobby and Chintu be x , y , and z .

Considering $x+y+z = 100 \cdot p$.

Given Anil's share was 70 percent = $70 \cdot p$.

As per the information provided :

His share of profit decreases by ₹ 420 if the overall profit goes down from 18% to 15%.

Since the profits are distributed in the ratio of their investments :

With a 3% decrease in the profits the value of profit earned by A decreased by Rs 420 which was 70 percent of the total invested.

$$\text{Hence for all three of them would be combinedly losing } (420) \cdot \left(\frac{10}{7}\right) = 600$$

Hence 3 percent profit was equivalent to Rs 600.

The initial investment is equivalent to Rs 20000.

This is the total amount invested.

Chintu's profit share increased by Rs 80 when the profit percentage increased by 2 %. A 2 percent increase in profit is equivalent to $\text{Rs } 20000 \cdot \frac{2}{100} = \text{Rs } 400$.

Of which Rs 80 is earned by Chintu which is 20% of the total Rs 400.

Hence he invested 20% of the total amount.

Bobby invested the other 10 percent.

$$10 \text{ percent of Rs } 20000 = \text{Rs } 2000$$

 VIDEO SOLUTION

32. A

Given,

$$C1 : C2 : C3 = 9 : 10 : 8 \dots i$$

$$C2 : C4 : C5 = 18 : 19 : 20 \dots ii$$

Let's multiply i by 9 and ii by 5

$$C1 : C2 : C3 = 81 : 90 : 72$$

$$C2 : C4 : C5 = 90 : 95 : 100$$

Therefore, $C1 : C2 : C3 : C4 : C5 = 81 : 90 : 72 : 95 : 100$

Given,

$$100x - 81x = 19$$

$$x = 1$$

Hence, total profit = $100 + 95 + 72 + 90 + 81 = 438$

 VIDEO SOLUTION

33. A

Let the number of dozens of candies he bought of each variety be x

$$\text{Hence total cost} = 12x + 15x = 27x$$

$$\text{Total selling price} = 16.50 \times 2x = 33x$$

$$\text{Profit} = 33x - 27x = 6x$$

$$\text{Given } 6x = 150 \Rightarrow x = 25$$

Hence he bought 50 dozens of candies in total

 VIDEO SOLUTION

34. 160

Total syrup - 110 kg

Total juice - 120 kg

It is given, cost price of syrup is 20% less than the cost price of juice.

Let the cost price of juice per kg be $10CP$

Cost price of syrup per kg is $8CP$

$$10\text{kg syrup} \rightarrow \text{cost price} = 80CP$$

It is given, 10kg syrup is sold at 10% profit. This implies selling price = $1.1 \times 80CP = 88CP$

$$20\text{kg juice} \rightarrow \text{cost price} = 200CP$$

It is given, 20kg juice is sold at 20% profit. This implies selling price = $1.2 \times 200CP = 240CP$

It is given, Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg

$$\text{Selling price of the remaining mixture} = 308.32 \times 200 = \text{Rs } 61664$$

$$\text{Total S.P} = 61664 + 328CP$$

$$\text{Total C.P} = 880CP + 1200CP = 2080CP$$

Overall profit = 64%

$$61664 + 328CP = \frac{164}{100} (2080CP)$$

Solving, we get $CP = 20$

$$\text{Cost price for syrup per kg} = 8CP = 8 \times 20 = \text{Rs } 160$$

 VIDEO SOLUTION

35. C

Let us assume the initial selling prices of A and B is p .

Given, she made profit of 20% on A $\Rightarrow 1.2 \times c = p \Rightarrow c = \frac{5p}{6} \Rightarrow$ cost of A is $\frac{5}{6}p$

Given, she made a loss of 10% on B $\Rightarrow 0.9 \times c = p \Rightarrow c = \frac{10p}{9} \Rightarrow$ cost of B is $\frac{10}{9}p$

Now, she sold them at a price such that a 10% profit is made on B

$$\Rightarrow \text{Selling price} = s = 11/10 * 10/9 p \Rightarrow \frac{11}{9}p$$

$$\Rightarrow \text{Profit \% on A} = \frac{\left(\frac{11}{9} - \frac{5}{6}\right)}{\left(\frac{5}{6}\right)} \times 100 = 46.66\% = \text{nearly } 47\%$$

[VIDEO SOLUTION](#)

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36. B

Let us fix the Cost Price of the product to be X, and the Selling Price of the product to be 1.4X, since it is given that it is fixed to have a profit of 40%.

If the CP has been 40% less, making the CP 0.6X,

And the selling price is 5 rupees less, making it 1.4X-5

Profit will be 50%,

$$\text{So, } 1.5(0.6X) = 1.4X - 5$$

$$0.9X = 1.4X - 5$$

$$0.5X = 5$$

$$X = 10$$

Original selling price will be 14.

[VIDEO SOLUTION](#)

37. D

We are told that, 10000 is deposited in bank A for a certain number of years at a simple interest of 5% per annum.

Let us say that the number of years is x

Total value of the deposit after x years is, $10000(1 + x(0.05))$

On maturity, the total amount received is deposited in bank B for another 5 years at a simple interest of 6% per annum

Here we know the years and the interest rate,

$$10000(1 + x(0.05))(1 + 5(0.06))$$

$$10000(1 + (0.05)x)(1.3)$$

Interest received from Bank A is $(x(0.05))10000$

Interest received from Bank B is $0.3(10000(1 + x(0.05)))$

This ratio is given to be 10:13.

$$\frac{x(0.05)}{0.3(1 + x(0.05))} = \frac{10}{13}$$

$$0.65x = 3 + 0.15x$$

$$0.5x = 3$$

$$x = 6$$

Hence the number of years the money was invested in Bank A is 6 years.

[VIDEO SOLUTION](#)

38. 2160

Let the cost price of the item be C

We are given that Bina sells this at 19% loss or at $(1 - 0.19)C = 0.81C$ at 4860

This gives us the value of C at Rs. 6000

If Bina had sold this at 17% profit, the selling price would have been $1.17 \times 6000 = 7020$

So Shyam bought the product at 4860 and sold it to Hari at 7020

Giving the profit made by Shyam to be $7020 - 4860 = 2160$

Therefore, 2160 is the correct answer.

 VIDEO SOLUTION

39. D

Let's take the amount invested by Sunil to be X.

The amount received by Anil at the end of 6 years would be $22000 \left(1 + \frac{4}{2 \times 100}\right)^{6 \times 2} = 22000 (1.02)^{12}$

The amount received by Sunil at the end of 5 years would be $X (1.02)^{10}$

In the 6th year, Sunil invests this at a simple interest of 10%, giving him an interest of $X (1.02)^{10} \times 0.1$

Giving the total amount with him at the end of 6 years to be $X (1.02)^{10} \times (1 + 0.1)$

Equating the final amount with Sunil and Anil, we get:

$$X (1.02)^{10} \times (1.1) = 22000 (1.02)^{12}$$

$$X = \frac{22000(1.02)^2}{1.1} = 20808$$

Therefore, Option D is the correct answer.

 VIDEO SOLUTION

40. A

Let us say the cost price of an item is X

It is said that it is marked to make a profit of 20%.

That means it is marked at 1.2X

Ravi gets a 10% discount on the marked price,

$$0.9 (1.2X) = 1.08X$$

Saves 15 rupees, so $1.2X - 1.08X$

$$0.12X = 15$$

$$X = 125$$

Profit made by Gopi is $0.08(125) = 10$ rupees.

 VIDEO SOLUTION

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41. C

We are told that there was two successive increments in the salary, with the second increment percentage twice the first one. Total Increment was 187.5%.

Drawing up the equation

$$(1 + z) (1 + 2z) = 1.875$$

$$1 + 3z + 2z^2 = 1.875$$

$$2z^2 + 3z - 0.875$$

$$z = \frac{(-3 \pm \sqrt{9+8(0.875)})}{4}$$

$$z = \frac{(-3 \pm 4)}{4}$$

$$z = 0.25$$

Answer is 25%.

[VIDEO SOLUTION](#)

42. 407

Let the number of white shirts be m , and the number of blue shirts be n . Hence, the total cost of the shirts = $(1000m+1125n)$, and the number of shirts is $(m+n)$

The average price of the shirts is $\frac{1000m+1125n}{m+n}$. It is given that he set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10%.

Hence, the average selling price of the shirts = $\left(\frac{1000m+1125n}{m+n} \right) \times \frac{5}{4} \times \frac{9}{10} = \frac{9}{8} \left(\frac{1000m+1125n}{m+n} \right)$

The average profit of the shirts = $\frac{9}{8} \left(\frac{1000m+1125n}{m+n} \right) - \frac{1000m+1125n}{m+n} = \frac{1}{8} \left(\frac{1000m+1125n}{m+n} \right)$

The total profit of the shirts = $\frac{1}{8} \left(\frac{1000m+1125n}{m+n} \right) \times (m+n) = \frac{1}{8} (1000m + 1125n)$

Now, $\Rightarrow \frac{1}{8} (1000m + 1125n) = 51000$

$\Rightarrow 1000m + 1125n = 51000 \times 8 = 408000$

Now to get the maximum number of shirts, we need to minimize n (since the coefficient of n is greater than the coefficient of m), but it can't be zero. Therefore, m has to be maximum.

$$m = \frac{408000-1125n}{1000}$$

The maximum value of m such that m , and both are integers is $m = 399$, and $n = 8$ (by inspection)

Hence, the maximum number of shirts = $m+n = 399+8 = 407$

[VIDEO SOLUTION](#)

43. A

As from his original cost values, number of TV's were 60 and VCR's were 40

And now total selling price will be = $12200 \times 60 + 18300 \times 40 = 1464000$

Hence profit = selling price - cost price

$$= 1464000 - 1200000$$

$$= 264000 \text{ rs.}$$

[VIDEO SOLUTION](#)

44. C

It is given that the maximum cost is 12 lakhs

$1200000 \leq T \times 10000 + V \times 15000$ (Where T is number of TV's and V is number of VCR's)

and as we know $T + V = 100$

By solving the above two equation, we will get $v \leq 40$

We have to keep the no. of VCRs as maximum, as they fetch higher profit than TV, so

$$T = 60$$

$$V = 40$$

[VIDEO SOLUTION](#)

45. 20

Let the total investment be $15x$ and the no. of years required be T years

$$\frac{(3x \times 6 \times T)}{100} + \frac{(5x \times 10 \times T)}{100} + \frac{(7x \times 1 \times T)}{100} \geq 15x$$

$$\text{or, } \frac{75xT}{100} \geq 15x$$

$$\text{or, } T \geq 20$$

So minimum value of T is 20 years

[VIDEO SOLUTION](#)

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46. B

It is given that Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. It is also known that he repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year.

The total amount at the end of the first year is: $200000 \times \frac{104}{100} \times \frac{104}{100} = 216320$

He repays 10320 rupees at the end of the first year, which implies the amount that remains unpaid at the end of the first year is 206000 rupees.

This unpaid amount will accrue interest for another two years.

Hence, the final amount at the end of three years is $206000 \times \frac{104}{100} \times \frac{104}{100} \times \frac{104}{100} = 240990.86$

Hence, the accrued interest in these two years is $(240990.86 - 206000) = 34990.86$ rupees.

Hence, the total interest accrued over the three years = $(34990.86 + 16320) = 51311$ rupees.

The correct option is B

[VIDEO SOLUTION](#)

47. D

Let the retail price be M and cost price be C.

Given,

$$0.85 M = 1.02 C$$

$$M = 1.2 C$$

If he wants 20% profit he has to sell at 1.2C, which is nothing but the retail price.

[VIDEO SOLUTION](#)

48. 9

Assuming the amount invested in the ratio 2:1 was 200x and 100x, then the fixed deposit investment = $1500000 - 300x$

Hence, the interest = $200x \times 4/100 = 8x$ and $100x \times 3/100 = 3x$

Interest from the fixed deposit = $(1500000 - 300x) \times 6/100 = 90000 - 18x$

Hence the total interest = $90000 - 18x + 8x + 3x = 90000 - 7x = 76000$

$$\Rightarrow 7x = 14000 \Rightarrow x = 2000$$

Hence, the fixed deposit investment = $1500000 - 300 \times 2000 = 900000 = 9$ lakhs

[VIDEO SOLUTION](#)

49. A

The total property consists of cash, gold coins and silver bars.

And ghosh babu gave equal parts to 4 daughters, hence they should have 25% of total property each.

As eldest daughter possess gold coins as 20% worth of total property, so 25000 cash should be equal to 5% of total property.

So total property will be $= \frac{25000 \times 100}{5} = 500000$

 VIDEO SOLUTION

50. D

Cost of 20 units = $240 + 20b + 400c$

Cost of 40 units = $240 + 40b + 1600c = \frac{5}{3} * (240 + 20b + 400c) \Rightarrow 720 + 120b + 4800c = 1200 + 100b + 2000c$

$\Rightarrow 480 = 20b + 2800c \Rightarrow 120 = 5b + 700c$

Cost of 60 units = $240 + 60b + 3600c = \frac{3}{2} (240 + 40b + 1600c) \Rightarrow 480 + 120b + 7200c = 720 + 120b + 4800c$

$\Rightarrow 240 = 2400c \Rightarrow c = 1/10$ and $b = 10$

Let the number of items needed for max profit be k

$CP = 240 + 10k + \frac{k^2}{10}$

$SP = 30k$

Profit = $SP - CP = 30k - 240 - 10k - \frac{k^2}{10} = 20k - 240 - \frac{k^2}{10}$

Maximum when $20 - \frac{k}{5} = 0$ or $k = 100$

Profit = $2000 - 240 - 1000 = 760$

 VIDEO SOLUTION

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51. 90000

For two years the compound interest is $\frac{PR(1)}{100} + \frac{PR(1)}{100} \left(1 + \frac{PR(1)}{100}\right)$

For three years the simple interest is $\frac{9PR}{100}$

Now $R(1) = 5\%$ and $R = 3\%$

Hence $\frac{5P}{100} + \frac{5P}{100} (1.05) - \frac{9P}{100} = 1125$

$\frac{-4P}{100} + \frac{5.25P}{100} = 1125$

$\frac{1.25P}{100} = 1125$

Solving we get $P = 90000$

 VIDEO SOLUTION

52. C

Assuming the cost price of pen = $100p$ and the cost price of book = $100b$

So, on selling a pen at 5% loss and a book at 15% gain, net gain = $-5p + 15b = 7 \dots 1$

On selling the pen at 5% gain and the book at 10% gain, net gain = $5p + 10b = 13 \dots 2$

Adding 1 and 2 we get, $25b = 20$

Hence $100b = 20 \times 4 = 80$,

C is the answer.

 VIDEO SOLUTION

53. A

Let the price of the painting be P

One cycle of price increase and decrease reduces the price by $x^2/100 * P = 441$

Let the new price be N $\Rightarrow P - x^2/100 * P = N$

Price after the second cycle = $N - x^2/100 * N = 1944.81$

$$\Rightarrow (P - x^2/100 * P)(1 - x^2/100) = 1944.81$$

$$\Rightarrow (P - 441)(1 - 441/P) = 1944.81$$

$$\Rightarrow P - 441 - 441 + 441^2/P = 1944.81$$

$$\Rightarrow P^2 - (882 + 1944.81)P + 441^2 = 0$$

$$\Rightarrow P^2 - 2826.81P + 441^2 = 0$$

From the options, the value 2756.25 satisfies the equation.

So, the price of the article is Rs 2756.25

 VIDEO SOLUTION

54. C

Let the principal amount be P and the interest rate be r.

$$\text{Then } P(1+r)^2 - P(1+r) = 806.25 \text{ -(1)}$$

$$P(1+r)^3 - P(1+r)^2 = 866.72 \text{ -(2)}$$

Dividing (2) by (1), we get:

$$\frac{(P(1+r)^3 - P(1+r)^2)}{P(1+r)^2 - P(1+r)} = \frac{866.72}{806.25}$$

$$\frac{((1+r)^2 - 1 - r)}{1+r-1} = 1.075$$

$$\frac{r^2+r}{r} = 1.075$$

$$r = 0.075 \text{ or } 7.5\%$$

$$\frac{(\text{Interest accrued in 4th yr})}{(\text{Interest accrued in 3rd yr})} = \frac{X}{866.72}$$

$$\frac{(P(1+r)^4 - P(1+r)^3)}{P(1+r)^3 - P(1+r)^2} = \frac{X}{866.72}$$

Dividing numerator and denominator by $P(1+r)^2$

$$\frac{r^2+2r+1-1-r}{1+r-1} = \frac{X}{866.72}$$

$$r+1 = \frac{X}{866.72}$$

$$X = 1.075 \times 866.72 = 931.72$$

 VIDEO SOLUTION

55. B

Bank A: 6% p.a. 1/2 yearly (CI)

Bank B: x% p.a (SI)

Bank C: 2x% p.a (SI)

Let Raju invest Rs P in bank B for t years. Hence, Rupa invests Rs 10,000 in bank C for 2t years.

Now,

$$P\left(\frac{x}{100}\right)t = P\left(1 + \frac{3}{100}\right)^2 - P$$

$$\left(\frac{x}{100}\right)t = 1.0609 - 1$$

$$\left(\frac{x}{100}\right)t = 0.0609$$

We need to calculate

$$SI = 10000 \times 2t \times \left(\frac{2x}{100}\right) = 40000 \left(\frac{x}{100}\right) t = 40000 \times 0.0609 = 2436$$

 VIDEO SOLUTION

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56. **16000**

Given,

Rate of interest = 10%

Since it is compounded half-yearly, $R=5\%$

$n=3$

We know, $A = P \left(1 + \frac{R}{100}\right)^n$

$$18522 = P (1 + 0.05)^3$$

$$\Rightarrow P = 16000$$

 VIDEO SOLUTION

57. **B**

Cost price per watch = 150

Cost price for 1500 watches = $1500 \times 150 = 225000$

Total expense = $225000 + 30000 = 255000$

Selling price for season = $1200 \times 250 = 300000$

For out of season = $300 \times 100 = 30000$

Total selling = $300000 + 30000 = 330000$

Profit = $330000 - 255000 = 75000$

 VIDEO SOLUTION

58. **B**

Let the manufacturing price of the table = x

Hence the price at which the wholesaler bought from the manufacturer = $1.1 \times x$

The price at which the retailer bought from the wholesaler = $1.3 \times 1.1 \times x$

The price at which the customer bought from the retailer = $1.5 \times 1.3 \times 1.1 \times x$

$$1.5 \times 1.3 \times 1.1 \times x = 4290$$

$$\Rightarrow x = 2000$$

 VIDEO SOLUTION

59. **70000**

Let his total savings be $100x$.

He invests $50x$ in fixed deposits. 30% of $50x$, which is $15x$ is invested in stocks and $35x$ goes to savings bank.

It is given $85x = 59500$

$$x = 700$$

Hence, $100x = 70000$

 VIDEO SOLUTION

60. **C**

let's say price of maruti car is x rs.
Sales = y
revenue = xy
Changed price = $1.3x$
changed value of sales = $0.8y$
new revenue = $1.04xy$
Percentage change in revenue = 4%

 [VIEW SOLUTION](#)

CAT Percentile Predictor

61. C

The cost price of the sunglass for Meenu when he purchased it for the first time was 1000 rupees, and he sold it to Kanu at 20% profit. Hence, the selling price of the sunglass is 1200 rupees, which Kanu purchased. Hence, the profit made by Meenu is $(1200 - 1000) = 200$ rupees.

Hence, the cost price of the same sunglass for Kanu is 1200 rupees, and now he sold it to Meenu at a 20% loss. Hence, the selling price of the sunglass now is $(1200 \times 0.8) = 960$ rupees.

The cost price of the same sunglass for Meenu when he purchased it for the second time was 960 rupees. Now Meenu sold it Tanu, at a certain price such that the total profit of Meenu becomes 500 rupees.

Hence, on the second transaction (selling it to Tanu), Meenu made a profit of $(500 - 200) = 300$ rupees.

Hence, the profit made by Minu in the second transaction is $(300/960) \times 100\% = 31.25\%$

The correct option is C

 [VIDEO SOLUTION](#)

62. B

Let a , b and c be the percentages of amount invested in options A, B and C respectively $\Rightarrow a + b + c = 100$

Return attained if there is a rise in the stock market $\Rightarrow 0.001a + 0.05b - 0.025c$

Return attained if there is a fall in the stock market $\Rightarrow 0.001a - 0.03b + 0.02c$

Maximum guaranteed return is attained when both are equal because it is indifferent to rise and fall in the market.

$$0.001a + 0.05b - 0.025c = 0.001a - 0.03b + 0.02c$$

$$\Rightarrow 0.08b = 0.045c \Rightarrow 16b = 9c$$

Let's put the values for a , b and c that satisfy the above equation.

$$b = 9, c = 16, a = 75 \Rightarrow \text{return} = 0.125$$

$$b = 18, c = 32, a = 50 \Rightarrow \text{return} = 0.15$$

$$b = 27, c = 48, a = 25 \Rightarrow \text{return} = 0.175$$

$$b = 36, c = 64, a = 0 \Rightarrow \text{return} = 0.2$$

Hence, the maximum guaranteed return is 0.2% and it is attained when 36% is invested in option B and 64% is invested in option C.

 [VIDEO SOLUTION](#)

63. D

Let the number of gold bars be $7a$ and the number of silver bars be $27a$.

The total value of the gold and silver bars is $500000 - 225000 = 275000$

Therefore, $4000 \times 7a + 1000 \times 27a = 275000$

$$\text{Or, } 55000a = 275000$$

$$\text{Or, } a = 5$$

Therefore the number of silver bars with Ghosh Babu is $27a = 135$

 VIDEO SOLUTION

64. B

The total property consists of cash, gold coins and silver bars.

And ghosh babu gave equal parts to 4 daughters, hence they should have 25% of total property each.

As eldest daughter possess gold coins as 20% worth of total property, so 25000 cash should be equal to 5% of total property.

$$\text{So total property will be} = \frac{25000 \times 100}{5} = 500000$$

Hence property amounting only gold coins and silver bars will be = Total property - Total Cash

$$\text{i.e.} = 500000 - (25000 + 50000 + 75000 + 75000)$$

$$= 2,75,000$$

 VIDEO SOLUTION

65. B

As we know for maximizing profit, Total Revenue = Marginal Cost

$$\text{i.e. } T \times 10000 + V \times 15000 = 1200000$$

$$\text{and } T + V = 120$$

By solving above two equations we will get

$$T = 120$$

$$V = 0$$

Hence ratio will be 0.

 VIDEO SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

66. C

Let's say total money was x rs.

So cost price of 40 mango will be = x ;

Hence cost price of 20 mangoes will be = $\frac{x}{2}$

$$\text{Taxi fare} = \frac{10x}{100}$$

$$\text{Total expense} = \frac{x}{2} + \frac{10x}{100} = \frac{6x}{10}$$

$$\text{Remaining money} = \frac{4x}{10}$$

Cost price of 1 orange will be = $\frac{x}{50}$

Hence in $\frac{4x}{10}$ rs. 20 oranges can be purchased.

 VIDEO SOLUTION

67. B

He earns $x\%$ on first 2000 and $y\%$ on rest of his income.

So on 4000 rs. , he will earn as follows:

$$2000 \frac{x}{100} + 2000 \frac{y}{100} = 700$$

$$\text{Or } x + y = 35$$

Similarly on 5000 rs. ,he will earn 900 as follows:

$$2000 \frac{x}{100} + 3000 \frac{y}{100} = 900$$

$$\text{Or } 20x + 30y = 900$$

On solving above equations, we will get value of $x = 15$

 VIDEO SOLUTION

68. **D**

Let's say he buy fruits of weights 3 kg., 4kg., 5 kg.

$$\text{Total kilograms of dry fruits} = 3 + 4 + 5 = 12$$

$$\text{Overall cost price} = 3 \cdot 100 + 4 \cdot 80 + 5 \cdot 60 = 300 + 320 + 300 = 920$$

$$\text{So cost price per kg.} = \frac{300 + 320 + 300}{12} = \frac{920}{12}$$

$$\text{Selling price} = \frac{920}{12} \times \frac{3}{2} = 115 \text{ per kg (Since Profit is 50\%)}$$

Hence answer will be D.

 VIDEO SOLUTION

69. **A**

Let's say cost price is 100

$$\text{gain} = 14.28$$

$$\text{selling price} = 114.28$$

Marked price = x (say)

$$\text{So } x - \frac{11.11x}{100} = \frac{8x}{9} = 114.28$$

$$\text{Or } x = 128.52$$

So marked price is 28.52% more than cost price.

 VIDEO SOLUTION

70. **A**

As shopkeeper gave 3 one-rupee change for 20 rs. change, Buyer must have ordered for a total of 17 rs. stamps.

Now buyer ordered for at least more than 1 stamp for each type

Hence the minimum he bought was:

$$2 \text{ stamp for 5 rupees} = 10 \text{ rs.}$$

$$2 \text{ stamp for 2 rupees} = 4 \text{ rs.}$$

$$2 \text{ stamp for 1 rupee} = 2 \text{ rs.}$$

For the total to be seventeen, the buyer must have purchased 3 one rupee stamps.

And 3 one rupee stamps were also there as changes given by shopkeeper.

$$\text{So total number of stamps} = 2+2+(3+3) = 10$$

 VIDEO SOLUTION

[Data Interpretation for CAT Questions \(download pdf\)](#)

71. **C**

Let the original weight of the diamond be equal to $10k$. So, after breaking into 4 pieces, the parts of the diamond weight $k, 2k, 3k, 4k$

The price of the diamond varies directly in proportion to the weight. Let us assume, the $P = C * W^2$ where C is a constant and W is the weight of the diamond.

Therefore, the original price is $C * 10k * 10k = 100k^2 * C$

The new weight is $Ck^2 + C(2k)^2 + C(3k)^2 + C(4k)^2 = 30k^2C$

The decrease in the price equals 70,000. So, $100k^2C - 30k^2C = 70000$

Or, $k^2C = 1000$

Therefore the original price = $100k^2C = 100000$

 VIDEO SOLUTION

72. 121000

We have to equate the installments and the amount due either at the time of borrowing or at the time when the entire loan is repaid. Let us bring all values to the time frame in which all the dues get settled, i.e., by the end of 2 years.

John borrowed Rs. 2,10,000 from the bank at 10% per annum. This loan will amount to $2,10,000 * 1.1 * 1.1 = \text{Rs. } 2,54,100$ by the end of 2 years.

Let the amount paid as installment every year be Rs.x.

John would pay the first installment by the end of the first year. Therefore, we have to calculate the interest on this amount from the end of the first year to the end of the second year. The loan will get settled the moment the second installment is paid.

$$\Rightarrow 1.1x + x = 2,54,100$$

$$2.1x = 2,54,100$$

$$\Rightarrow x = \text{Rs. } 1,21,000.$$

Therefore, 121000 is the correct answer.

 VIDEO SOLUTION

73. C

It is given that a merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him.

Hence, the cost price of 105 cm clothes is 100 rupees.

It is also known that he marked the price of 100 cm clothes as 110 rupees, and gave a 5% discount, and he cheated his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers.

Hence, the selling price of 95 cm clothes is $110 * (19/20)$ rupees.

Therefore, the selling price of 105 cm clothes is 115.5 rupees.

Hence, the profit is 15.5%

The correct option is C

 VIDEO SOLUTION

74. B

Break even implies that cost price is equal to selling price
Hence let's say in season x watches were sold
Cost price will be = $1500 \times 150 + 30000 = 255000$
So total selling price = $250x + (1500 - x)100 = 255000$
Or $x = 700$

 VIDEO SOLUTION

75. A

Let the savings invested in first part and second part be 'x' and 'y', respectively.

It is given,

$$\frac{x \times 15 \times 4}{100} = \frac{y \times 12 \times 3}{100}$$

$$60x = 36y$$

$$5x = 3y$$

$$\text{Required percentage} = \frac{x}{x+y} \times 100 = \frac{3}{3+5} \times 100 = 37.5\%$$

The answer is option A.

 VIDEO SOLUTION

Logical Reasoning for CAT Questions (download pdf)

76. B

Let the total production cost be 100.

Hence, selling price is 120.

Price of German component A is 30 and the price of the US component B is 50

After change in exchange rate, price of German component is $30 \times 1.2 = 36$

and price of US component is $50 \times 0.88 = 44$

Total increase equals $(36+44)-(30+50) = 0$

Hence, the total production cost did not change.

As the selling price also did not change, the gain percentage equals 20%

 VIEW SOLUTION

77. 9

Let us assume the amount invested to be P , and the rate of interest to be r .

Value of the investment after 3 years will be, $P(1+r)^3$

Value of the investment after 5 years will be, $P(1+r)^5$

$$(1+r)^2 = \frac{36}{25}$$

$$(1+r)^2 = 1.44$$

$$r = 0.2$$

We need to find the value of n for which $4000(1+r)^n > 20000$

$$(1+r)^n > 5$$

$$(1.2)^n > 5$$

We see that,

$$1.2^8 = 4.2999$$

$$1.2^9 = 5.15$$

Hence it takes 9 years to grow to over 20,000.

 VIDEO SOLUTION